Signal Processing and Detection

AI-driven Approach

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A Spectrum Sensing Approach

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Preamble

THIS textbook explores contemporary advancements in signal processing and detection, with a focus on the transition toward intelligent, efficient, and interdisciplinary methodologies. Propelled by rapid computational progress and the increasing complexity of real-world applications, these emerging paradigms challenge conventional signal theory frameworks, fostering innovative, adaptive, and robust solutions across diverse domains.

A central development is the emergence of intelligent signal processing, which integrates sophisticated computational techniques—particularly artificial intelligence and machine learning. These methodologies enhance the analysis, interpretation, and exploitation of signals through adaptive, data-driven approaches. They prove especially transformative in applications demanding high accuracy and automation, including signal detection and classification in multi-user multiple-input multiple-output (MU-MIMO) systems, anomaly detection in cybersecurity data streams, and performance optimization in high-dimensional or uncertain environments where traditional deterministic models fall short.

Equally significant is the advent of sparse signal processing, which leverages the intrinsic sparsity of signals to enable accurate reconstruction from far fewer samples than required by the classical Nyquist–Shannon sampling theorem. Techniques such as compressive sensing have profound implications across multiple fields. In medical imaging, they enable substantial reductions in scan times, improving patient comfort and operational efficiency. Within the Internet of Things (IoT) and next-generation wireless networks (e.g., 5G and 6G), sparse signal processing optimizes bandwidth utilization, facilitating efficient data transmission in resource-constrained settings.

By synthesizing these trends, this book provides a comprehensive framework for understanding the theoretical foundations and practical applications of modern signal processing and detection. It bridges the gap between traditional signal theory and cutting-edge interdisciplinary approaches, offering valuable insights for researchers, engineers, and practitioners navigating the evolving landscape of signal science.

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Contents

Glossary

Acronym	Description
WGN	White Gaussian Noise
PSD	Power Spectral Density
FT	Fourier Transform
STFT	Short-Time Fourier Transform
ACF	Autocorrelation Function
PDF	Probability Density Function
HMM	Hidden Markov Model
MSE	Mean Square Error
NNs	Neural Networks
MLP	Multi-Layer Perceptron
$\mathbb{E}\{\cdot\}$	Expectation operator (mean value)
$\mathcal{L}_p$-norm	Euclidean p-norm
$\langle \cdot, \cdot \rangle$	Inner product
$\| \cdot \|_2$	$\mathcal{L}_2$-norm (Euclidean norm)
$R_\eta(\tau)$	Autocorrelation function (ACF) of η
$m_{n\xi}$	n-order initial moment of ξ
$K_x(\tau)$	Covariance function of x
$S_\xi(\omega)$	Power spectral density of ξ
$p(x)$	Probability of outcome x
$I(x_i)$	Self-information of outcome x_i
$H(X)$	Entropy: expected self-information
$\mathcal{N}_\xi\,(,)$	Standard Gaussian distribution of ξ
$\rho(\tau)$	Correlation coefficient at lag τ
$\mathcal{C}^*$	Cost function:
LTI systems	Linear Time-Invariant systems
$H(f)$	Transfer function:
BNN	Bayesian Neural Networks
VAE	Variational Autoencoders
MCMC	Markov Chain Monte Carlo
KL divergence	Kullback-Leibler divergence
ELBO	Evidence Lower Bound

1 Signal Representation

1 Random Signals and Noise

1.1 Stationary Random Signals

Random signals [Sequence view]: Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. A (real or complex) random signal (or stochastic process) is a collection of random variables $\{\xi(s){:}s{\in}S\}$, where each $\xi(s)$ is $(\mathcal{F}, \mathcal{B}(\Xi))$-measurable, so that

$$\xi(s) : \Omega \longrightarrow \Xi.$$

Here, S is the index set (domain) and Ξ is the state space (codomain), typically $\mathbb{R}$ or $\mathbb{C}$ with the Borel σ-algebra $\mathcal{B}(\Xi)$. If S is continuous (e.g., $\mathbb{R}$), we speak of a *continuous-time* random signal; if $S{=}\{s_k\}_{k\in\mathbb{Z}}$ is discrete, we speak of a *discrete-time random sequence.*

SINCE random signals are predominantly indexed by time $t{\in}\mathbb{R}$, they are referred to as time-dependent processes. Consequently, phenomena evolving over a time-related argument and described by a random signal $\xi(t)$ can be analyzed through a longitudinal collection of measurements or recorded values, denoted as $\{\xi_m(t){\in}\Xi{:}m{\in}M\}$, as illustrated in Figure 1.1. In this context, each m-th realization $\xi_m(t) = \xi(t, \omega_m)$ corresponds to evaluating the process at a specific outcome $\omega_m \in \Omega$, representing one observed trajectory. The complete set of all trajectories $\{\xi_m(t)\}$ recorded over the observation interval constitutes the *ensemble.* These trajectories correspond to the same underlying random phenomenon, $\xi(t)$.

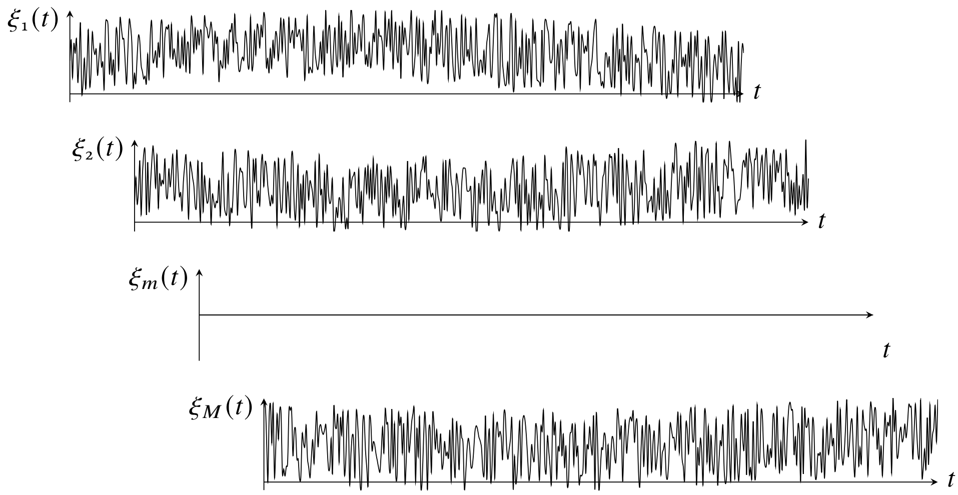


Figure 1.1: Ensemble of random signals. Each m-th realization $\xi_m(t)$ corresponds to evaluating $\xi(t, \cdot)$ at a specific outcome $\omega_m \in \Omega$, representing one observed trajectory.

Equivalently, a *discrete-time* stochastic process can be represented within the framework of a measure-preserving dynamical system. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathscr{T} : \Omega \longrightarrow \Omega$ be a measurable, measure-preserving transformation, i.e.,

$$\mathbb{P}\big(\mathscr{T}^{-1}(F)\big) = \mathbb{P}(F), \qquad \forall F \in \mathcal{F}.$$

Given an observable $X_0 \colon (\Omega, \mathcal{F}) \longrightarrow (\Xi, \mathcal{B}(\Xi))$, we define the process

$$X_n(\omega) = X_0(\mathscr{T}^n\omega), \qquad n \in \mathbb{Z}_{\geq 0}.$$

Then $(X_n)_{n\geq 0}$ is strictly stationary. For bilateral processes indexed by $n \in \mathbb{Z}$, we additionally require $\mathscr{T}$ to be invertible and both $\mathscr{T}$ and $\mathscr{T}^{-1}$ to be measure-preserving. The distribution of the process is thereby induced by $(\Omega, \mathcal{F}, \mathbb{P}, \mathscr{T})$ and the observable X_0.

For continuous-time stationary processes, the analogous construction uses a measurable measure-preserving flow $\{\mathscr{T}_t\}_{t\in\mathbb{R}}$ and defines $X(t,\omega) = X_0(\mathscr{T}_t\omega)$ for $t \in \mathbb{R}$.

The sequence view emphasizes finite-dimensional distributions and is the starting point for Kolmogorov's extension theorem, while the dynamical systems view highlights temporal evolution, where properties of $\mathscr{T}$ translate into statistical regularities of the process.

From a sequence view, random signals are analyzed by examining how their randomness structure varies over time or other domains. These structural changes can be evaluated through the finite-dimensional distributions (FDDs[1]) of the process, i.e., the joint distributions of $\xi(s_1), \ldots, \xi(s_k)$ for any finite collection $\{s_1, \ldots, s_k\} \subset S$, which fully characterize the probabilistic structure of ξ.

Strict stationarity. A random signal $\{\xi(t){:}t\in\mathbb{R}\}$ is *strictly stationary* (or *strong-sense stationary*) if for every $T\in\mathbb{N}$, every $(t_1, \ldots, t_T) \in \mathbb{R}^T$, and every time shift $\tau \in \mathbb{R}$, the finite-dimensional distributions satisfy

$$(\xi(t_1), \ldots, \xi(t_T)) \overset{d}{=} (\xi(t_1+\tau), \ldots, \xi(t_T+\tau)),$$

where $\overset{d}{=}$ denotes equality in distribution. Equivalently, the joint CDFs satisfy:

$$F_{\xi(t_1),\ldots,\xi(t_T)}(x_1, \ldots, x_T) = F_{\xi(t_1+\tau),\ldots,\xi(t_T+\tau)}(x_1, \ldots, x_T).$$

Complementing this distributional description, statistical moments provide concise descriptors of a signal's stochastic structure. The joint PDF $p(\xi(t_1), \ldots, \xi(t_T))$ governs their temporal evolution and yields the marginal distributions $\{p(\xi(t_i))\}_{i=1}^{T}$, from which moments at each instant t_i are computed.

Thus, given a collection of M observed realizations $\{\xi_m(t_i) : m = 1, \ldots, M\}$, two different averaging operations characterize the statistical properties of the observed signal.

i) *Ensemble average.* At a fixed t_i, the n-th *ensemble-averaged* moment performed across realizations is

$$\mathbb{E}\left\{\xi_m^n(t_i)\right\} = \int_{\Omega} \left(\xi(t_i, \omega)\right)^n d\mathbb{P}(\omega). \tag{1.1}$$

where $\mathbb{E}\{\cdot\}$ denotes the expectation with respect to $(\Omega, \mathcal{F}, \mathbb{P})$.

If M realizations are available, the ensemble moment can be estimated empirically as

$$\widehat{\mathbb{E}\{\xi^n(t_i)\}} = \frac{1}{M}\sum_{m=1}^{M} \xi_m^n(t_i). \tag{1.2}$$

ii) *Time average.* For the m-th realization $\xi_m(t)$, the n-th time-averaged moment is computed

[1] These FDDs must satisfy Kolmogorov consistency conditions: symmetry under permutations and compatibility under marginalization.

along a single trajectory. Whenever the limit exists,

$$\overline{\{\xi_m^n\}} = \lim_{T\to\infty} \int_{-\infty}^{\infty} \xi_m^n(t)\, w_T(t)\, dt, \tag{1.3}$$

where $w_T : \mathbb{R} \longrightarrow \mathbb{R}$ is a normalized weighting function satisfying

$$\int_{-\infty}^{\infty} w_T(t)\, dt = 1, \quad w_T(t) \geq 0$$

$$\text{Particularly, } w_T(t) = \begin{cases} \dfrac{1}{T}, & 0 \leq t \leq T, \\ 0, & \text{otherwise.} \end{cases} \tag{1.4a}$$

$$\text{Hence, } \quad \overline{\{\xi_m^n\}} = \lim_{T\to\infty} \frac{1}{T} \int_0^T \xi_m^n(t)\, dt. \tag{1.4b}$$

When second moments are finite, each sample value $\xi(t)$ can be treated not only as a scalar random variable but also as an element of a $\mathcal{L}_2$ space of random variables. This viewpoint is useful because it allows distances, orthogonality, and projections to be defined in terms of mean-square error.

Random Hilbert space. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a complete probability space. We define the space of square-integrable complex-valued random variables as

$$\mathcal{H} = \mathcal{L}_2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C}) = \{\zeta : \Omega \longrightarrow \mathbb{C} \mid \zeta \text{ is } \mathcal{F}\text{-measurable and } \mathbb{E}\{|\zeta|^2\} < \infty\}.$$

Strictly speaking, this space identifies random variables that are equal $\mathbb{P}$-almost surely; its elements are therefore equivalence classes of square-integrable random variables.
The space $\mathcal{H}$ becomes a Hilbert space when equipped with the sesquilinear inner product

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \mathbb{E}\{\xi\, \overline{\eta}\}, \qquad \xi, \eta \in \mathcal{H},$$

where $\overline{\eta}$ denotes the complex conjugate of η. The associated norm is

$$\|\xi\|_{\mathcal{H}} = \sqrt{\langle \xi, \xi \rangle_{\mathcal{H}}} = \left(\mathbb{E}\{|\xi|^2\}\right)^{1/2},$$

which corresponds to the RMS value of the random variable ξ. The induced metric is

$$d_{\mathcal{H}}(\xi, \eta) = \|\xi - \eta\|_{\mathcal{H}} = \left(\mathbb{E}\{|\xi - \eta|^2\}\right)^{1/2}, \qquad \xi, \eta \in \mathcal{H}.$$

Under this metric, $(\mathcal{H}, d_{\mathcal{H}})$ is complete by the Riesz–Fischer theorem. Hence, $\mathcal{H}$ is a Hilbert space. The Hilbert-space structure allows regularity, approximation, and orthogonality of random signals to be described through mean-square distance.

Let $\{\xi(t){:}t{\in}S\}$ be a random signal such that $\xi(t){\in}\mathcal{H},\ \forall t{\in}S$. Assume that $S{\subseteq}\mathbb{R}$ and that t_0 is an interior point of S. The process is said to be *mean-square continuous* at $t_0{\in}S$ if

$$\lim_{\tau\to 0} d_{\mathcal{H}}\big(\xi(t_0+\tau), \xi(t_0)\big) = \lim_{\tau\to 0} \left(\mathbb{E}\{|\xi(t_0+\tau) - \xi(t_0)|^2\}\right)^{1/2} = 0. \tag{1.5}$$

That is, $\lim_{\tau\to 0} \|\xi(t_0+\tau) - \xi(t_0)\|_{\mathcal{H}} = 0$.

Thus, mean-square continuity means that the mapping $t \mapsto \xi(t)$ is continuous as a function from the time domain S into the Hilbert space $\mathcal{H}$. In other words, nearby time instants correspond to nearby random variables in the mean-square metric.

Mean-square continuity implies continuity in probability because convergence in $\mathcal{L}_2$ implies convergence in probability. Its relation with almost-sure continuity requires additional assumptions: almost-sure continuity alone does not automatically imply mean-square continuity unless suitable integrability or domination conditions are also satisfied. For this reason, mean-square continuity is especially useful in the analysis of second-order random processes, where second moments are central.

The next example contrasts the Fourier case with nonorthogonal functions whose overlap depends on the reference probability measure.

Example: Fourier kernels in $\mathcal{L}_2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C})$
Let $\Omega = [0, 1]$, $\mathcal{F} = \mathcal{B}([0, 1])$, and $\mathbb{P} = \lambda$, where λ denotes the Lebesgue probability measure on $[0, 1]$. Define the complex-valued random variables

$$\xi(\omega) = e^{jk2\pi\omega}, \qquad \eta(\omega) = e^{jl2\pi\omega}, \qquad k, l \in \mathbb{Z}, \quad \omega \in [0, 1].$$

These represent the k-th and l-th Fourier basis functions on the unit interval.
Square integrability. Since $|\xi(\omega)| = |e^{jk2\pi\omega}| = 1$ and $|\eta(\omega)| = |e^{jl2\pi\omega}| = 1$ for all $\omega \in [0, 1]$ and all $k, l \in \mathbb{Z}$,

$$\mathbb{E}\{|\xi|^2\} = \int_0^1 1\, d\lambda(\omega) = 1 < \infty, \qquad \mathbb{E}\{|\eta|^2\} = \int_0^1 1\, d\lambda(\omega) = 1 < \infty,$$

hence $\xi, \eta \in \mathcal{H} = \mathcal{L}_2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{C})$ for all $k, l \in \mathbb{Z}$.
Inner product. With $\langle \xi, \eta \rangle_{\mathcal{H}} = \mathbb{E}\{\xi\overline{\eta}\}$,

$$\begin{aligned}\langle \xi, \eta \rangle_{\mathcal{H}} &= \int_0^1 e^{jk2\pi\omega} \cdot \overline{e^{jl2\pi\omega}}\, d\lambda(\omega) = \int_0^1 e^{jk2\pi\omega} \cdot e^{-jl2\pi\omega}\, d\omega \\ &= \int_0^1 e^{j2\pi(k-l)\omega}\, d\omega.\end{aligned}$$

We distinguish two cases:

- If $k = l$:

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \int_0^1 e^0\, d\omega = \int_0^1 1\, d\omega = 1.$$

- If $k \neq l$:

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \int_0^1 e^{j2\pi(k-l)\omega}\, d\omega = \left.\frac{e^{j2\pi(k-l)\omega}}{j2\pi(k-l)}\right|_0^1 = \frac{e^{j2\pi(k-l)} - 1}{j2\pi(k-l)} = \frac{1-1}{j2\pi(k-l)} = 0.$$

Thus, we have the *orthonormality relation*:

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \delta_{kl} = \begin{cases} 1, & k = l, \\ 0, & k \neq l, \end{cases}$$

where δ_{kl} is the Kronecker delta.

Norms. By orthonormality,

$$\|\xi\|_{\mathcal{H}} = \sqrt{\mathbb{E}\{|\xi|^2\}} = \sqrt{\langle \xi, \xi\rangle_{\mathcal{H}}} = \sqrt{1} = 1,$$

and similarly $\|\eta\|_{\mathcal{H}} = 1$ for all $k, l \in \mathbb{Z}$.
Induced distance. Using the identity $|e^{j\theta} - e^{j\phi}|^2 = 2 - 2\cos(\theta - \phi)$, we obtain

$$\begin{aligned} d_{\mathcal{H}}(\xi,\eta)^2 &= \mathbb{E}\{|\xi - \eta|^2\} \\ &= \int_0^1 |e^{jk2\pi\omega} - e^{jl2\pi\omega}|^2 \, d\lambda(\omega) = \int_0^1 \left|e^{jk2\pi\omega}\right|^2 \left|1 - e^{j(l-k)2\pi\omega}\right|^2 d\omega \\ &= \int_0^1 \left|1 - e^{j2\pi(l-k)\omega}\right|^2 d\omega \\ &= \int_0^1 \big(2 - 2\cos(2\pi(l-k)\omega)\big)\, d\omega. \end{aligned}$$

Again, two cases arise:

$$\begin{aligned} &\text{if } k = l: \quad d_{\mathcal{H}}(\xi,\eta)^2 = \int_0^1 0\, d\omega = 0 \quad \Rightarrow \quad d_{\mathcal{H}}(\xi,\eta) = 0. \\ &\text{if } k \neq l: \quad d_{\mathcal{H}}(\xi,\eta)^2 \\ &\qquad = \int_0^1 \big(2 - 2\cos(2\pi(l-k)\omega)\big)\, d\omega \\ &\qquad = 2\omega\Big|_0^1 - \frac{2\sin(2\pi(l-k)\omega)}{2\pi(l-k)}\Big|_0^1 = 2 - \frac{2\sin(2\pi(l-k))}{2\pi(l-k)} \\ &\qquad = 2 - 0 = 2, \end{aligned}$$

since $\sin(2\pi n) = 0$ for all $n \in \mathbb{Z}$. Thus $d_{\mathcal{H}}(\xi,\eta) = \sqrt{2}$.
Consequently, the distance between distinct Fourier modes is constant:

$$d_{\mathcal{H}}(\xi,\eta) = \begin{cases} 0, & k = l, \\ \sqrt{2}, & k \neq l. \end{cases}$$

This shows that $\{e^{jk2\pi\omega}\}_{k\in\mathbb{Z}}$ forms an *orthonormal system* in $\mathcal{L}_2([0,1], \mathcal{B}, \lambda; \mathbb{C})$, with all distinct elements equidistant.

The next example contrasts the Fourier case with nonorthogonal functions whose overlap depends on the reference probability measure.

Example: Gaussian kernels in $\mathcal{H}$ Let $\Omega = \mathbb{R}$, $\mathcal{F} = \mathcal{B}(\mathbb{R})$, and let $\mathbb{P}$ be the standard Gaussian probability measure defined by

$$\mathbb{P}(A) = \int_A \phi(\omega)\, d\omega, \quad A \in \mathcal{B}(\mathbb{R}), \qquad \phi(\omega) = \frac{1}{\sqrt{2\pi}} e^{-\omega^2/2}.$$

Define the real-valued random variables on $\mathcal{H} = \mathcal{L}_2(\Omega, \mathcal{F}, \mathbb{P}; \mathbb{R})$ as a pair of Gaussian kernels centered at μ_1 and μ_2, respectively:

$$\xi(\omega) = \exp\left(-\frac{(\omega-\mu_1)^2}{2\sigma^2}\right), \quad \eta(\omega) = \exp\left(-\frac{(\omega-\mu_2)^2}{2\sigma^2}\right), \quad \omega \in \mathbb{R},\ \sigma > 0,\ \mu_1, \mu_2 \in \mathbb{R}.$$

Square integrability. Since $0 < \xi(\omega) \leq 1$ and $0 < \eta(\omega) \leq 1$ for all $\omega \in \mathbb{R}$, we have

$$|\xi(\omega)|^2 \leq 1, \qquad |\eta(\omega)|^2 \leq 1.$$

$$\text{Hence} \quad \mathbb{E}\left\{|\xi|^2\right\} = \int_{\mathbb{R}} e^{-(\omega-\mu_1)^2/\sigma^2}\, \phi(\omega)\, d\omega < \infty, \qquad \mathbb{E}\left\{|\eta|^2\right\} < \infty.$$

Therefore $\xi, \eta \in \mathcal{H}$.
Inner product. With $\langle \xi, \eta \rangle_{\mathcal{H}} = \mathbb{E}\{\xi\eta\}$, we obtain

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \int_{\mathbb{R}} \exp\left(-\frac{(\omega-\mu_1)^2}{2\sigma^2}\right) \exp\left(-\frac{(\omega-\mu_2)^2}{2\sigma^2}\right) \phi(\omega)\, d\omega.$$

By completing the square,

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \frac{\sigma}{\sqrt{\sigma^2+2}} \exp\left(\frac{2\mu_1\mu_2 - (\sigma^2+1)(\mu_1^2+\mu_2^2)}{2\sigma^2(\sigma^2+2)}\right).$$

In particular, $\langle \xi, \eta \rangle_{\mathcal{H}} > 0$, so these kernels are generally not orthogonal.
Norms. Taking $\mu_1 = \mu_2$ in the previous formula gives

$$\|\xi\|_{\mathcal{H}}^2 = \langle \xi, \xi \rangle_{\mathcal{H}} = \frac{\sigma}{\sqrt{\sigma^2+2}} \exp\left(-\frac{\mu_1^2}{\sigma^2+2}\right),$$

$$\text{and similarly} \quad \|\eta\|_{\mathcal{H}}^2 = \frac{\sigma}{\sqrt{\sigma^2+2}} \exp\left(-\frac{\mu_2^2}{\sigma^2+2}\right).$$

Note that $\|\xi\|_{\mathcal{H}}^2$ is a decreasing function of $|\mu_1|$ for fixed σ: as the kernel center moves away from the origin, the Gaussian measure $\mathbb{P}$ assigns less mass to the region where ξ is large, thereby reducing its norm in $\mathcal{H}$.
Induced distance. The Hilbert-space distance satisfies

$$d_{\mathcal{H}}(\xi, \eta)^2 = \|\xi - \eta\|_{\mathcal{H}}^2 = \|\xi\|_{\mathcal{H}}^2 + \|\eta\|_{\mathcal{H}}^2 - 2\langle \xi, \eta \rangle_{\mathcal{H}}.$$

Therefore,

$$d_{\mathcal{H}}(\xi, \eta)^2 = \frac{\sigma}{\sqrt{\sigma^2+2}} \left(e^{-\mu_1^2/(\sigma^2+2)} + e^{-\mu_2^2/(\sigma^2+2)} - 2\exp\left(\frac{2\mu_1\mu_2 - (\sigma^2+1)(\mu_1^2+\mu_2^2)}{2\sigma^2(\sigma^2+2)}\right)\right).$$

The distance $d_{\mathcal{H}}(\xi, \eta)$ vanishes if and only if $\xi = \eta$ $\mathbb{P}$-a.s., which in this parametrization is equivalent to $\mu_1 = \mu_2$.
Unlike the Fourier modes on $[0, 1]$, the distance between these Gaussian kernels is not translation invariant. It depends not only on the separation between the centers μ_1 and μ_2, but also on their location relative to the reference probability measure $\mathbb{P}$. Centers located far from the origin may have small $\mathcal{H}$-norms because the standard Gaussian measure assigns little probability mass to those regions.
Therefore, Gaussian kernels define elements of $\mathcal{H}$, and the Hilbert-space inner product measures their probability-weighted overlap under the reference measure $\mathbb{P}$.

67

Kernel Expansion of Random Signals

The previous examples show that functions can be treated as elements of a Hilbert space and compared through an inner product. This viewpoint also allows one to approximate a signal by a finite weighted combination of kernels. Let $\xi(t)$ be a square-integrable random signal and choose Gaussian kernels

$$\mathcal{N}_k(t) = \exp\left(-\frac{(t-\mu_k)^2}{2\sigma_k^2}\right), \qquad k = 1, \ldots, K.$$

where $\mu_k \in \mathbb{R}$ denotes the center of the k-th kernel and $\sigma_k > 0$ controls its width. Note: σ_k here denotes kernel width, distinct from process variance σ_ξ^2.

A finite Gaussian-kernel approximation of $\xi(t)$ is

$$\hat{\xi}(t) = \sum_{k=1}^{K} c_k \mathcal{N}_k(t).$$

$$\text{or in matrix form, for sampled data,} \quad \hat{\boldsymbol{\xi}} \approx \boldsymbol{\mathcal{N}} \boldsymbol{c},$$

where the k-th column of $\boldsymbol{\mathcal{N}}$ contains the sampled values of $\phi_k(t)$, and the entries of $\boldsymbol{c}$ are the coefficients c_k.

The coefficients c_k are chosen so that $\hat{\xi}(t)$ is close to $\xi(t)$ in the selected norm. For example, in the least-squares sense, they minimize

$$\boldsymbol{c}_{opt} = \arg\min_{\boldsymbol{c}} \|\boldsymbol{\xi} - \boldsymbol{\mathcal{N}}\boldsymbol{c}\|^2 .$$

where the deterministic matrix $\boldsymbol{\mathcal{N}}$ describes the time-domain kernel structure, while the random coefficient vector $\boldsymbol{c}_{opt}$ describes how strongly each kernel contributes for each realization of the signal. Because Gaussian kernels are generally not orthogonal, the coefficients are not given by independent projections. Instead, they are coupled through the Gram matrix of the kernels.

This formulation connects random-signal expansions with kernel methods and prepares the transition toward neural models, where the fixed kernel expansion is replaced by a learned nonlinear representation.

Pytask: RBF Network for Noisy Signal Reconstruction.
Objective: Use a radial basis function (RBF) network to approximate a noisy sinusoidal signal.
Task Steps:

1. Generate training data:
   - Signal: $x(t) = \cos(2\pi f_0 t) + \cos(6\pi f_0 t)/3 + \cos(10\pi f_0 t)/5$ with $f_0 = 5$ Hz
   - Add Gaussian noise: $y(t) = x(t) + \eta(t), \eta \sim \mathcal{N}(0, \sigma^2)$
   - Sample at $N = 100$ points over $t \in [0, 2]$ seconds
2. Design RBF layer:
   - Use $K = 10$ Gaussian kernels with centers uniformly spaced in $[0, 2]$
   - Set all widths to $\sigma_k = 0.2$
   - Compute activation matrix $\boldsymbol{\mathcal{N}} \in \mathbb{R}^{100\times10}$
3. Train output weights:

    - Solve $\boldsymbol{c}_\lambda = (\mathcal{N}^\top \mathcal{N} + \lambda \boldsymbol{I})^{-1} \mathcal{N}^\top \boldsymbol{y}$
    - Test $\lambda \in \{0, 0.01, 0.1, 1\}$

4. Evaluate:
    - Plot: noisy data, true signal, RBF approximation for each λ
    - Compute MSE: $\frac{1}{N} \|\boldsymbol{x} - \{\hat{x}\}\|^2$
    - Identify optimal λ via cross-validation

Expected Output:

- Figure with 4 subplots (one per λ value)
- Table of MSE vs. λ
- Discussion: How does λ affect overfitting?

Extensions:

1. Compare with polynomial regression (degree 5, 10)
2. Experiment with kernel placement (uniform vs. k-means)
3. Implement a two-layer RBF network with learnable centers

89 *Reference:* `01a3KernelRBF.ipynb`

1.2 Correlation Function

(Orthogonality and correlation): Two random variables $\xi, \eta \in \mathcal{H}$ are said to be *orthogonal* if

$$\langle \xi, \eta \rangle_{\mathcal{H}} = \mathbb{E}\left\{\xi\, \overline{\eta}\right\} = 0.$$

Thus, the inner product structure of $\mathcal{H}$ provides a geometric notion of perpendicularity between random variables.
The corresponding centered second-order quantity is the covariance-type inner product

$$\mathbb{E}\left\{\left(\xi - \mathbb{E}\left\{\xi\right\}\right)\overline{\left(\eta - \mathbb{E}\left\{\eta\right\}\right)}\right\}.$$

The random variables ξ and η are said to be *uncorrelated* if

$$\mathbb{E}\left\{\left(\xi - \mathbb{E}\left\{\xi\right\}\right)\overline{\left(\eta - \mathbb{E}\left\{\eta\right\}\right)}\right\} = 0.$$

Equivalently, $\mathbb{E}\left\{\xi\, \overline{\eta}\right\} = \mathbb{E}\left\{\xi\right\}\, \overline{\mathbb{E}\left\{\eta\right\}}$.

For zero-mean random variables, that is, $\mathbb{E}\left\{\xi\right\} = 0$, $\mathbb{E}\left\{\eta\right\} = 0$, the centered and non-centered inner products coincide:

$$\mathbb{E}\left\{\left(\xi - \mathbb{E}\left\{\xi\right\}\right)\overline{\left(\eta - \mathbb{E}\left\{\eta\right\}\right)}\right\} = \mathbb{E}\left\{\xi\, \overline{\eta}\right\}.$$

Therefore, for zero-mean random variables, orthogonality is equivalent to uncorrelatedness.

For nonzero-mean random variables, orthogonality and uncorrelatedness are generally different conditions. Orthogonality requires $\mathbb{E}\left\{\xi\, \overline{\eta}\right\} = 0$, whereas uncorrelatedness requires

$$\mathbb{E}\left\{\xi\, \overline{\eta}\right\} = \mathbb{E}\left\{\xi\right\}\, \overline{\mathbb{E}\left\{\eta\right\}}.$$

That is, *Statistical Independence* is strictly stronger than *Uncorrelatedness*, with the hierarchy:

$$\text{Independence} \longrightarrow \text{Uncorrelatedness} \longrightarrow \text{Orthogonality (for zero-mean)}.$$

A random signal $\xi(t) \in \mathcal{H}$ is said to be *second-order* if its first- and second-order ensemble moments are finite. In particular, for every $t \in T$, $\mathbb{E}\left\{|\xi(t)|^2\right\} < \infty$. Equivalently, $\xi(t) \in \mathcal{H}$, $\forall t \in T$.

Let $w : T \longrightarrow \mathbb{C}$ be a deterministic window function with $w \in \mathcal{L}_1(T) \cap \mathcal{L}_2(T)$, and assume that the following integrals are well defined. Then the windowed random variable

$$Y_w = \int_T w(t)\xi(t)\, dt$$

$$\text{has mean } \mathbb{E}\left\{Y_w\right\} = \mathbb{E}\left\{\int_T w(t)\xi(t)\, dt\right\} = \int_T w(t)\mu_\xi(t)\, dt, \tag{1.6}$$

where $\mu_\xi(t) = \mathbb{E}\left\{\xi(t)\right\}$. The above interchange of $\mathbb{E}\left\{\cdot\right\}$ and $\int$ follows from the Fubini–Tonelli, provided $\int_T w(t)\mathbb{E}\left\{|\xi(t)|\right\} dt < \infty$.

Moreover, using the autocorrelation function $R_\xi(t_1, t_2) = \mathbb{E}\left\{\xi(t_1)\xi^*(t_2)\right\}$, we obtain

$$\begin{aligned}
\mathbb{E}\left\{|Y_w|^2\right\} &= \mathbb{E}\left\{\left|\int_T w(t)\xi(t)\,dt\right|^2\right\} \\
&= \int_T\int_T w(t_1)w^*(t_2)R_\xi(t_1, t_2)\,dt_1 dt_2 < \infty.
\end{aligned} \tag{1.7}$$

Equivalently, using the covariance function

$$K_\xi(t_1, t_2) = \mathbb{E}\left\{\big(\xi(t_1) - \mu_\xi(t_1)\big)\big(\xi(t_2) - \mu_\xi(t_2)\big)^*\right\},$$

the variance of Y_w is

$$\mathbb{E}\left\{|Y_w - \mathbb{E}\left\{Y_w\right\}|^2\right\} = \int_T\int_T w(t_1)w^*(t_2)K_\xi(t_1, t_2)\,dt_1 dt_2. \tag{1.8}$$

Therefore,

$$\mathbb{E}\left\{|Y_w|^2\right\} = \int_T\int_T w(t_1)w^*(t_2)K_\xi(t_1, t_2)\,dt_1 dt_2 + \left|\int_T w(t)\mu_\xi(t)\,dt\right|^2. \tag{1.9}$$

Here, w is a deterministic window that ensures the integrals are well defined and allows the interchange of $\mathbb{E}\left\{\cdot\right\}$ and $\int$ under standard integrability conditions ($\mathcal{L}_1$ ensures the mean integral converges; $\mathcal{L}_2$ ensures Fubini applies in the second-moment computation.). The notation $(\cdot)^*$ denotes complex conjugation.

Consequently, the second-order ensemble moments of random signals are defined as follows:

$$\mu_\xi(t) = \mathbb{E}\left\{\xi(t)\right\}, \tag{1.10a}$$

$$\sigma_\xi^2(t) = \mathbb{E}\left\{|\xi(t) - \mu_\xi(t)|^2\right\}, \tag{1.10b}$$

$$K_{\xi\eta}(t_m, t_n) = \mathbb{E}\left\{\big(\xi(t_m) - \mu_\xi(t_m)\big)\overline{\big(\eta(t_n) - \mu_\eta(t_n)\big)}\right\}, \tag{1.10c}$$

$$R_{\xi\eta}(t_m, t_n) = \mathbb{E}\left\{\xi(t_m)\eta^*(t_n)\right\}. \tag{1.10d}$$

Here, $K_{\xi\eta}$ is the cross-covariance function, whereas $R_{\xi\eta}$ is the cross-correlation function. For $\xi = \eta$, $R_{\xi\xi}$ becomes the ACF and $K_{\xi\xi}$ becomes the autocovariance function.

The non-centered pairwise second-order moment

$$R_{\xi\eta}(t_m, t_n) = \mathbb{E}\left\{\xi(t_m)\eta(t_n)\right\}$$

is called the *cross-correlation function.* When the same random signal is used in both arguments, $\xi = \eta$, it becomes the *autocorrelation function* (ACF),

$$R_\xi(t_m, t_n) = R_{\xi\xi}(t_m, t_n) = \mathbb{E}\left\{\xi(t_m)\xi(t_n)\right\}.$$

The centered pairwise second-order moment

$$K_{\xi\eta}(t_m, t_n) = \mathbb{E}\left\{\big(\xi(t_m) - \mu_\xi(t_m)\big)\big(\eta(t_n) - \mu_\eta(t_n)\big)\right\}$$

is called the *cross-covariance function*. When $\xi{=}\eta$, it becomes the *autocovariance function*,

$$K_{\xi}(t_m, t_n) = K_{\xi\xi}(t_m, t_n).$$

Since the present subsection considers real-valued random signals, complex conjugates are omitted. For complex-valued random signals, the second argument in correlation and covariance expressions must be conjugated.

Consider now the sum of two second-order real-valued random signals,

$$\eta(t) = \xi_1(t) + \xi_2(t).$$

Its mean is $\mu_\eta(t) = \mathbb{E}\left\{\eta(t)\right\} = \mu_{\xi_1}(t) + \mu_{\xi_2}(t).$

The autocovariance function of $\eta(t)$ is

$$\begin{aligned} K_\eta(t_1, t_2) &= \mathbb{E}\left\{\big(\eta(t_1) - \mu_\eta(t_1)\big)\big(\eta(t_2) - \mu_\eta(t_2)\big)\right\} \\ &= \mathbb{E}\left\{\Big[\big(\xi_1(t_1) - \mu_{\xi_1}(t_1)\big) + \big(\xi_2(t_1) - \mu_{\xi_2}(t_1)\big)\Big]\Big[\big(\xi_1(t_2) - \mu_{\xi_1}(t_2)\big) + \big(\xi_2(t_2) - \mu_{\xi_2}(t_2)\big)\Big]\right\} \\ &= K_{\xi_1}(t_1, t_2) + K_{\xi_1\xi_2}(t_1, t_2) + K_{\xi_2\xi_1}(t_1, t_2) + K_{\xi_2}(t_1, t_2). \end{aligned} \tag{1.11}$$

Equivalently, written term by term,

$$\begin{aligned} K_\eta(t_1, t_2) = {} & \mathbb{E}\left\{\big(\xi_1(t_1) - \mu_{\xi_1}(t_1)\big)\big(\xi_1(t_2) - \mu_{\xi_1}(t_2)\big)\right\} \\ & + \mathbb{E}\left\{\big(\xi_1(t_1) - \mu_{\xi_1}(t_1)\big)\big(\xi_2(t_2) - \mu_{\xi_2}(t_2)\big)\right\} \\ & + \mathbb{E}\left\{\big(\xi_2(t_1) - \mu_{\xi_2}(t_1)\big)\big(\xi_1(t_2) - \mu_{\xi_1}(t_2)\big)\right\} \\ & + \mathbb{E}\left\{\big(\xi_2(t_1) - \mu_{\xi_2}(t_1)\big)\big(\xi_2(t_2) - \mu_{\xi_2}(t_2)\big)\right\}. \end{aligned} \tag{1.12}$$

According to the nature of each term, the following cases can be considered:

Case 1: If $\xi_1(t)$ and $\xi_2(t)$ are mutually uncorrelated at the pair of instants (t_1, t_2), then

$$K_{\xi_1\xi_2}(t_1, t_2) = K_{\xi_2\xi_1}(t_1, t_2) = 0,$$

and therefore

$$K_\eta(t_1, t_2) = K_{\xi_1}(t_1, t_2) + K_{\xi_2}(t_1, t_2).$$

For the non-centered autocorrelation, if additionally

$$R_{\xi_1\xi_2}(t_1, t_2) = 0, \qquad R_{\xi_2\xi_1}(t_1, t_2) = 0,$$

then

$$R_\eta(t_1, t_2) = R_{\xi_1}(t_1, t_2) + R_{\xi_2}(t_1, t_2).$$

This stronger condition is automatically satisfied when the processes are zero-mean and mutually uncorrelated at the pair of instants (t_1, t_2).

Case 2: If one term is a random signal $\xi_1(t)$ and the other is a deterministic real-valued signal

$x(t)$, then

$$\eta(t) = \xi_1(t) + x(t).$$

The mean is

$$\mu_\eta(t) = \mathbb{E}\{\xi_1(t) + x(t)\} = \mu_{\xi_1}(t) + x(t).$$

Since the deterministic component is removed by centering, the covariance is unchanged:

$$K_\eta(t_1, t_2) = K_{\xi_1}(t_1, t_2).$$

However, the non-centered autocorrelation is not generally unchanged. It is

$$R_\eta(t_1, t_2) = R_{\xi_1}(t_1, t_2) + \mu_{\xi_1}(t_1)x(t_2) + x(t_1)\mu_{\xi_1}(t_2) + x(t_1)x(t_2).$$

Case 3: If the deterministic signal is a real-valued constant $x(t) = c$, then

$$\eta(t) = \xi_1(t) + c.$$

The mean is

$$\mu_\eta(t) = \mu_{\xi_1}(t) + c.$$

The covariance is still unchanged:

$$K_\eta(t_1, t_2) = K_{\xi_1}(t_1, t_2).$$

The non-centered autocorrelation becomes

$$R_\eta(t_1, t_2) = R_{\xi_1}(t_1, t_2) + \mu_{\xi_1}(t_1)c + c\,\mu_{\xi_1}(t_2) + c^2.$$

If $\xi_1(t)$ is zero-mean, this reduces to

$$R_\eta(t_1, t_2) = R_{\xi_1}(t_1, t_2) + c^2.$$

Example: Compute the ACF of a random signal composed of sinusoidal components with random amplitudes. Let

$$\xi_n(t) = \xi_{nc}\cos(\omega_n t) + \xi_{ns}\sin(\omega_n t), \quad \xi_{nc} \in \mathbb{R}, \xi_{ns} \in \mathbb{R}$$

where ξ_{nc} and ξ_{ns} are random coefficients. Assume $\mathbb{E}\{\xi_{nc}\} = \mathbb{E}\{\xi_{ns}\} = \mathbb{E}\{\xi_{nc}\xi_{ns}\} = 0$, and $\mathbb{E}\left\{\xi_{nc}^2\right\} = \sigma_{\xi_{nc}}^2$, $\mathbb{E}\left\{\xi_{ns}^2\right\} = \sigma_{\xi_{ns}}^2$.

Then the autocorrelation function of the n-th component is

$$\begin{aligned}
R_{\xi_n}(t_1, t_2) &= \mathbb{E}\left\{\xi_n(t_1)\xi_n(t_2)\right\} \\
&= \mathbb{E}\left\{(\xi_{nc}\cos(\omega_n t_1) + \xi_{ns}\sin(\omega_n t_1))\,(\xi_{nc}\cos(\omega_n t_2) + \xi_{ns}\sin(\omega_n t_2))\right\} \\
&= \mathbb{E}\left\{\xi_{nc}^2\right\}\cos(\omega_n t_1)\cos(\omega_n t_2) + \mathbb{E}\left\{\xi_{ns}^2\right\}\sin(\omega_n t_1)\sin(\omega_n t_2) \\
&\quad + \mathbb{E}\left\{\xi_{nc}\xi_{ns}\right\}\cos(\omega_n t_1)\sin(\omega_n t_2) + \mathbb{E}\left\{\xi_{ns}\xi_{nc}\right\}\sin(\omega_n t_1)\cos(\omega_n t_2) \\
&= \sigma_{\xi_{nc}}^2\cos(\omega_n t_1)\cos(\omega_n t_2) + \sigma_{\xi_{ns}}^2\sin(\omega_n t_1)\sin(\omega_n t_2).
\end{aligned} \tag{1.13}$$

If, in addition, $\sigma_{\xi_{nc}}^2 = \sigma_{\xi_{ns}}^2 = \sigma_{\xi_n}^2$, then

$$\begin{aligned}
R_{\xi_n}(t_1, t_2) &= \sigma_{\xi_n}^2\left[\cos(\omega_n t_1)\cos(\omega_n t_2) + \sin(\omega_n t_1)\sin(\omega_n t_2)\right] \\
&= \sigma_{\xi_n}^2\cos\big(\omega_n(t_1 - t_2)\big).
\end{aligned} \tag{1.14}$$

Therefore, under the equal-variance and uncorrelated-coefficient assumptions, the ACF depends only on the lag $t_1 - t_2$, so the component is wide-sense stationary.
Now consider the sum

$$\xi(t) = \sum_{n \in N} \xi_n(t).$$

If the components $\{\xi_n(t)\}_{n \in N}$ are mutually uncorrelated, then the ACF of $\xi(t)$ is

$$\begin{aligned}
R_{\xi}(t_1, t_2) &= \sum_{n \in N} R_{\xi_n}(t_1, t_2) \\
&= \sum_{n \in N} \sigma_{\xi_n}^2\cos\big(\omega_n(t_1 - t_2)\big).
\end{aligned} \tag{1.15}$$

142

Wide-sense stationarity. A second-order real-valued random signal $\xi(t)$ is said to be *wide-sense stationary* (WSS) if its mean is constant and its autocorrelation function depends only on the time lag. Formally,

$$\mu_{\xi}(t) = \mathbb{E}\left\{\xi(t)\right\} = \mu_{\xi}, \qquad \text{independent of } t, \tag{1.16a}$$

$$R_{\xi}(t_1, t_2) = \mathbb{E}\left\{\xi(t_1)\xi(t_2)\right\} = R_{\xi}(t_2 - t_1) = R_{\xi}(\tau), \qquad \tau = t_2 - t_1. \tag{1.16b}$$

Equivalently, since the mean is constant, the autocovariance function also depends only on the lag:

143

$$K_{\xi}(t_1, t_2) = \mathbb{E}\left\{\big(\xi(t_1) - \mu_{\xi}\big)\big(\xi(t_2) - \mu_{\xi}\big)\right\} = K_{\xi}(t_2 - t_1) = K_{\xi}(\tau).$$

Moreover,

$$K_{\xi}(\tau) = R_{\xi}(\tau) - \mu_{\xi}^2.$$

Thus, a WSS process has time-invariant first-order behavior and second-order dependence that is determined only by the separation between observation instants, not by their absolute location in time.

144 *Remark.* The WSS definition above is a second-order condition: it involves only the mean

and autocorrelation. It is therefore weaker than strict stationarity, which requires invariance of all finite-dimensional distributions.

Under the standard shift construction, a strictly stationary process can be represented as an observable evolving under a measure-preserving transformation. Conversely, if X_0 is an observable on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and $\mathcal{T} : \Omega \to \Omega$ is measure-preserving, then

$$X_n(\omega) = X_0(\mathcal{T}^n \omega), \qquad n \in \mathbb{Z}_{\geq 0},$$

defines a strictly stationary discrete-time process. In this qualified sense, the stochastic-process and dynamical-system viewpoints are connected through stationary processes.

Ergodicity of random signals. Let $\{\xi^{(k)}(t)\}_{k=1}^{M}$ be M realizations of a second-order wide-sense stationary process, each with a well-defined first moment. At a fixed instant t_i, the empirical ensemble mean across realizations is

$$\widehat{\mu}_\xi(t_i) = \frac{1}{M} \sum_{k=1}^{M} \xi^{(k)}(t_i).$$

For a single realization $r \in \{1, \ldots, M\}$, the time average over the observation interval $[0, T]$ is

$$\overline{\{\xi^{(r)}\}}_T = \frac{1}{T} \int_0^T \xi^{(r)}(t)\, dt.$$

If the signal is sampled at $t_j = j\Delta t$, $j = 0, \ldots, N_T - 1$, with $\Delta t = T/N_T$, then

$$\overline{\{\xi^{(r)}\}}_T \approx \frac{\Delta t}{T} \sum_{j=0}^{N_T-1} \xi^{(r)}(t_j) = \frac{1}{N_T} \sum_{j=0}^{N_T-1} \xi^{(r)}(t_j).$$

For an ergodic WSS process, when the number of samples N_T is sufficiently large, the time average converges to the ensemble mean:

$$\widetilde{\mu}_{\xi_k} \;=\; \mu_\xi(t_i), \tag{1.17}$$

almost surely for almost every realization r.

A WSS process is said to be *ergodic in the mean* if, for almost every realization r,

$$\lim_{T \to \infty} \overline{\{\xi^{(r)}\}}_T = \mathbb{E}\{\xi(t)\} = \mu_\xi.$$

Since the process is WSS, μ_ξ is independent of t. In practice, when only a finite observation interval is available, the time average $\overline{\{\xi^{(r)}\}}_T$ is used as an estimator of the ensemble mean μ_ξ.

More generally, ergodicity refers to the equality between ensemble averages and time averages for a given statistic. In engineering applications, stationarity is typically assumed as a prerequisite for ergodicity. Throughout this text, ergodicity is treated as a statistic-specific property of stationary processes—applicable to the mean, variance, autocorrelation, and other statistical measures.

For a WSS process that is ergodic in the relevant statistic, ensemble moments can be estimated

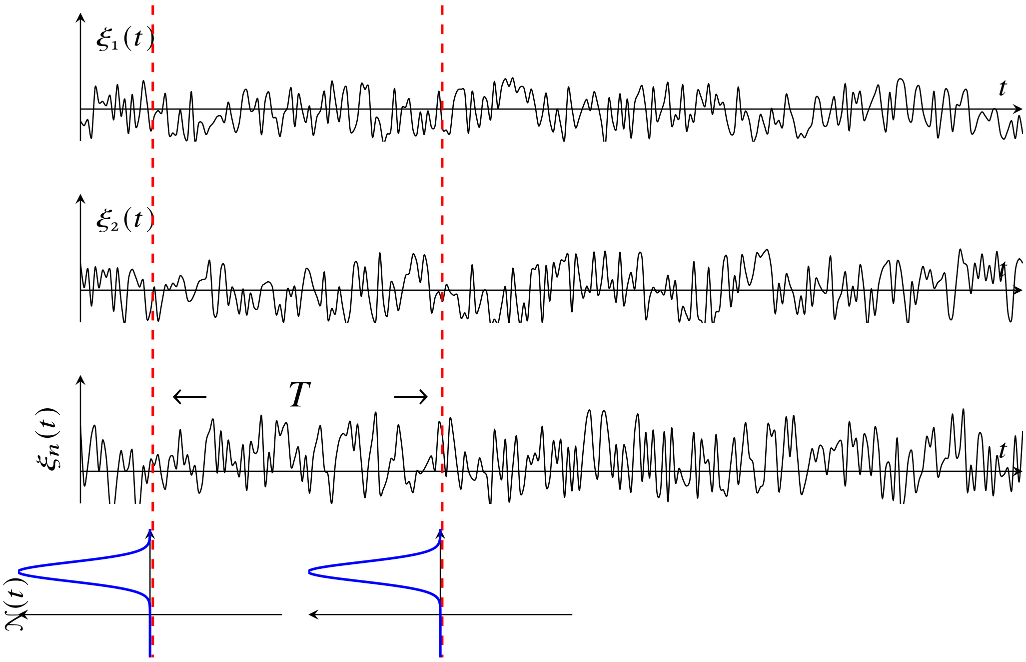


Figure 1.2: Ensemble of ergodic signals

from a single sufficiently long realization. Using the uniform weighting function

$$w_T(t) = \begin{cases} \frac{1}{T}, & 0 \le t \le T, \\ 0, & \text{otherwise,} \end{cases}$$

the mean is estimated by

$$\mu_\xi = \mathbb{E}\{\xi(t)\} = \lim_{T\to\infty} \frac{1}{T} \int_0^T \xi^{(r)}(t)\,dt = \lim_{T\to\infty} \overline{\{\xi^{(r)}\}}_T.$$

If the process is also ergodic in autocorrelation, then

$$\begin{aligned} R_\xi(\tau) &= \mathbb{E}\{\xi(t)\xi(t+\tau)\} \\ &= \lim_{T\to\infty} \frac{1}{T} \int_0^{T-\tau} \xi^{(r)}(t)\xi^{(r)}(t+\tau)\,dt, \quad \tau \ge 0. \end{aligned} \tag{1.18}$$

For negative lags, one may use the symmetry relation $R_\xi(-\tau) = R_\xi(\tau)$ for real-valued WSS processes. Equivalently, one may integrate over the valid overlap interval and let $T \to \infty$, so edge effects vanish asymptotically.

Similarly, the autocovariance can be estimated as

$$\begin{aligned} K_\xi(\tau) &= \mathbb{E}\left\{\big(\xi(t) - \mu_\xi\big)\big(\xi(t+\tau) - \mu_\xi\big)\right\} \\ &= \lim_{T\to\infty} \frac{1}{T} \int_0^{T-|\tau|} \big(\xi^{(r)}(t) - \mu_\xi\big)\big(\xi^{(r)}(t+\tau) - \mu_\xi\big)\,dt, \qquad \tau \in \mathbb{R}. \end{aligned}$$

For real-valued WSS processes,

$$K_\xi(\tau) = R_\xi(\tau) - \mu_\xi^2.$$

The moments of an ergodic WSS signal admit the following physical interpretation:

- The mean value represents the *constant component.*
- The squared mean value represents the *power of the constant component.*
- The mean-square value $\mathbb{E}\{\xi^2(t)\} = R_\xi(0)$ represents the *average power.*
- The variance $K_\xi(0)$ represents the *power of the alternating component.*
- The standard deviation $\sqrt{K_\xi(0)}$ relates to the *RMS value of the alternating component.*
- The ACF describes how signal values separated by a lag τ are linearly related.

The uniform weighting function $w_T(t)$ can also be interpreted as the impulse response of an averaging filter. Its Fourier transform, called here the *spectral characteristic*, is

$$\Phi_T(\omega) = \mathscr{F}\{w_T(t)\} = \int_{-\infty}^{\infty} w_T(t) e^{-j\omega t}\, dt.$$

For the rectangular averaging window,

$$\Phi_T(\omega) = e^{-j\omega T/2}\,\mathrm{sinc}\left(\frac{\omega T}{2}\right), \quad \mathrm{sinc}(x) = \frac{\sin x}{x}.$$

Pytask: Pytask

Generate through a prompt, step-by-step, the procedures described in Algorithm 1 for verifying the ergodicity condition over an ensemble of random signals (see Figure 1.3).

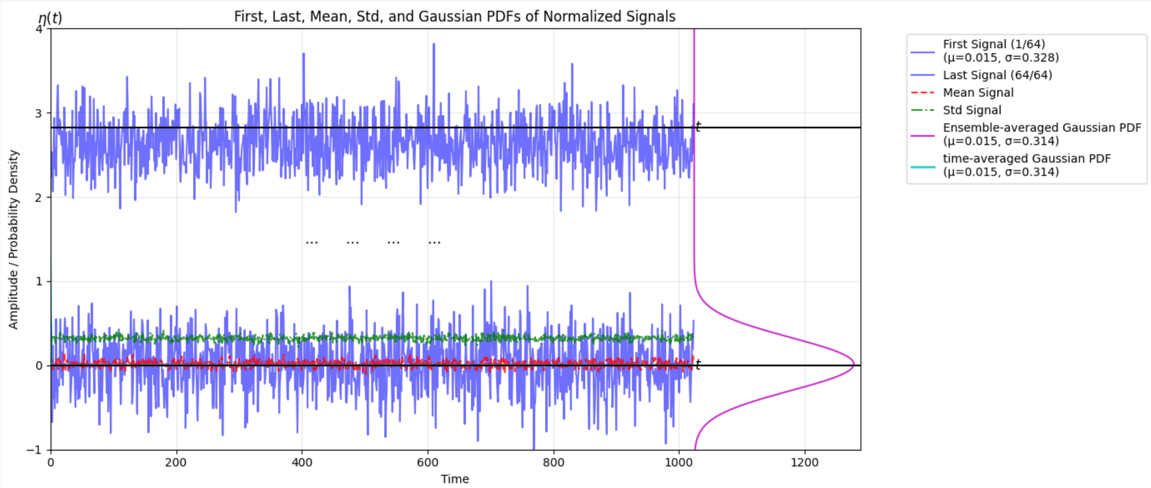


Figure 1.3: Illustration obtained according Algorithm 1. See 11aErgodic

Algorithm 1 Testing Ergodicity of random generated signals

Initialize:

$num\_samples \quad num\_signals \quad mean \quad std$

Step 1: Generate Two Random Signals

Generate $signalsi = \{1, 2\} \leftarrow$ random_normal$(mean, std, size = num_samples)$

Step 2: Normalize and Plot Two Signals

$signali\_normalized \leftarrow i \cdot \frac{(signali - mini)}{(maxi - mini)} - 1, \forall i$

Plot $signali\_normalized$, $i = 1, 64$ with label "Signal i",

Step 3: Add legends to Each Plot

Plot i: Add vertical line at $x = 0$, color black Add text "$x()$" at $(x = 0, y =$ max_ylim$)$, centered, bottom-aligned

Step 4: Generate 64 Signals and Plot First and Last

Generate $signals \leftarrow$random_normal$(mean, std, size = (num_signals, num_samples))$

Compute $mean\_signal$ and std_signal Across sample Points, plot both sets in bottom plot

Step 5:Plot Gaussian PDF Separately

$gaussian\_pdf \leftarrow \frac{1}{avg\_std\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{x\_pdf - avg\_mean}{avg\_std}\right)^2\right)$

Rotate and Place Gaussian PDF at the end of bottom plot

Step 6: Compute moments over time

Compute Mean and Std over time samples for each i Signal, $\forall i$

$mean\_i \leftarrow$ mean$(signal_i)$, $std\_i \leftarrow$ std$(signal_i)$

Compute Averages of Means and Stds over time

$avg\_mean \leftarrow$ mean$(mean_i)$ $avg\_std \leftarrow$ mean$(std_i), \forall i$

Step 7: Compute time Gaussian PDF

$time\_gaussian\_pdf \leftarrow \frac{1}{avg\_std\sqrt{2\pi}} \exp\left(-\frac{1}{2}\left(\frac{t - avg\_mean}{avg\_std}\right)^2\right)$

Superpose time Gaussian PDF over ensemble PDF

Plot $pdf\_x\_rotated, pdf\_y\_rotated$, label "Rotated Gaussian PDF", color magenta

Compute difference between time-averaged and ensemble averaged moments

Add legend/Display plot

Properties of the ACF.

Throughout this subsection, $\xi(t)$ is assumed to be a real-valued wide-sense stationary random signal. Therefore, complex conjugates are omitted, and the autocorrelation and autocovariance functions are written as

$$R_\xi(\tau) = \mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\}, \qquad K_\xi(\tau) = \mathbb{E}\left\{\left(\xi(t)-\mu_\xi\right)\left(\xi(t+\tau)-\mu_\xi\right)\right\}.$$

The following properties are highlighted.

(a). *Parity.*

For a real-valued WSS process, the autocorrelation and autocovariance functions are even functions of the lag.

Indeed, using stationarity,

$$R_\xi(\tau) = \mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\}.$$

Since multiplication of real numbers is commutative,

$$\mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\} = \mathbb{E}\left\{\xi(t+\tau)\xi(t)\right\}.$$

By WSS, the last expression corresponds to the correlation at lag $-\tau$. Hence,

$$R_\xi(\tau) = R_\xi(-\tau). \tag{1.19}$$

The same argument applied to the centered process $\xi(t)-\mu_\xi$ gives

$$K_\xi(\tau) = K_\xi(-\tau).$$

(b). *Maximum value.*

The autocorrelation and autocovariance functions satisfy the Cauchy–Schwarz bounds

$$\left|R_\xi(\tau)\right| \le R_\xi(0), \qquad \left|K_\xi(\tau)\right| \le K_\xi(0). \tag{1.20}$$

For the autocorrelation function, the Cauchy–Schwarz inequality for expectations gives

$$\left|\mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\}\right|^2 \le \mathbb{E}\left\{\xi^2(t)\right\}\mathbb{E}\left\{\xi^2(t+\tau)\right\}.$$

Since the process is WSS,

$$\mathbb{E}\left\{\xi^2(t)\right\} = \mathbb{E}\left\{\xi^2(t+\tau)\right\} = R_\xi(0).$$

Therefore, $\left|R_\xi(\tau)\right|^2 \le R_\xi^2(0)$, and hence $\left|R_\xi(\tau)\right| \le R_\xi(0)$. Similarly, applying Cauchy–Schwarz to the centered process $\xi(t)-\mu_\xi$ gives $\left|K_\xi(\tau)\right| \le K_\xi(0)$.

The values at the origin are

$$K_\xi(0) = \mathbb{E}\left\{\left(\xi(t)-\mu_\xi\right)^2\right\} = \sigma_\xi^2, \tag{1.21a}$$

$$R_\xi(0) = \mathbb{E}\left\{\xi^2(t)\right\}. \tag{1.21b}$$

Thus, $K_\xi(0)$ is the variance, while $R_\xi(0)$ is the mean-square value, or average power, of the process. In general,

$$R_\xi(0) = K_\xi(0) + \mu_\xi^2 = \sigma_\xi^2 + \mu_\xi^2.$$

(c). *Continuity.*

If $R_\xi(\tau)$ is continuous at $\tau = 0$, then $R_\xi(\tau)$ is uniformly continuous on $\mathbb{R}$. Similarly, if $K_\xi(\tau)$ is continuous at $\tau = 0$, then $K_\xi(\tau)$ is uniformly continuous on $\mathbb{R}$.

For a real-valued WSS process,

$$\mathbb{E}\left\{\big(\xi(t+\tau) - \xi(t)\big)^2\right\} = 2\big(R_\xi(0) - R_\xi(\tau)\big).$$

Hence, continuity of R_ξ at the origin is equivalent to mean-square continuity of the process. The uniform-continuity result follows from the positive semi-definiteness of valid autocorrelation functions.

(d). *Periodicity.*

If a WSS process has a periodic autocorrelation function with period $T_0 > 0$, then

$$R_\xi(\tau + T_0) = R_\xi(\tau), \qquad \forall \tau \in \mathbb{R}. \tag{1.22}$$

Similarly, if the autocovariance is periodic with the same period, then

$$K_\xi(\tau + T_0) = K_\xi(\tau).$$

Remark. Periodicity of the ACF does not imply that individual sample paths are periodic. For example, random-phase sinusoidal processes may have periodic ACFs while their individual realizations need not be periodic in the deterministic sense.

(e). *Positive semi-definiteness and spectral non-negativity.*

The autocorrelation function of a WSS process is positive semi-definite. That is, for any $n \in \mathbb{N}$, any time points $\{t_1, \ldots, t_n\}$, and any complex coefficients $\{c_1, \ldots, c_n\}$,

$$\sum_{i=1}^{n} \sum_{j=1}^{n} c_i c_j^* R_\xi(t_i - t_j) \geq 0. \tag{1.23}$$

Although $\xi(t)$ is real-valued in this subsection, complex coefficients are used for consistency with the general theory and Bochner's theorem.

By Bochner's theorem, R_ξ is positive semi-definite if and only if its FT is a non-negative measure. This measure is the power spectral distribution of the process. When $R_\xi \in \mathcal{L}_1(\mathbb{R})$, the power spectral density exists as an ordinary function and is given by

$$S_\xi(\omega) = \mathscr{F}\{R_\xi(\tau)\} = \int_{-\infty}^{\infty} R_\xi(\tau) e^{-j\omega\tau}\, d\tau \geq 0 \quad \text{for almost every } \omega \in \mathbb{R}. \tag{1.24}$$

With this angular-frequency convention, the inverse relation is

$$R_\xi(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_\xi(\omega) e^{j\omega\tau}\, d\omega.$$

Thus, spectral non-negativity follows from the positive semi-definiteness of R_ξ, not merely from the even symmetry of the ACF.

Remark. Any valid PSD estimate should be non-negative. Negative values usually indicate numerical or estimation artifacts, such as insufficient data length, inconsistent windowing, or an invalid autocorrelation estimate.

(f). *Asymptotic decorrelation.*

A WSS process is said to be *asymptotically uncorrelated* if

$$\lim_{|\tau|\to\infty} K_\xi(\tau) = 0. \tag{1.25}$$

Since

$$R_\xi(\tau) = K_\xi(\tau) + \mu_\xi^2,$$

it follows that

$$\lim_{|\tau|\to\infty} R_\xi(\tau) = \mu_\xi^2. \tag{1.26}$$

For zero-mean processes, this reduces to

$$\lim_{|\tau|\to\infty} R_\xi(\tau) = 0.$$

Remark. Asymptotic decorrelation and ergodicity are independent properties; neither implies the other in general. Asymptotic decorrelation must be verified separately, often through mixing conditions or decay estimates of temporal dependencies.

245 *Correlation Interval*:

- *Threshold Method* The correlation interval τ_c is the smallest positive value of τ for which the normalized ACF falls below a threshold $\beta \ll 1$:

$$\beta = \frac{R_\xi(\tau_c)}{\sigma_\xi^2} \approx 0.05 \text{ to } 0.1. \tag{1.27}$$

  Typically, $\beta = 1/e \approx 0.368$ (for exponential ACFs) or $\beta = 0.1$ (practical threshold).

- *Integral Method* The *equivalent correlation interval* $\Delta\tau_e$ is defined as half the width of a rectangle with unit height and area equal to the normalized ACF:

$$\Delta\tau_e = \frac{1}{R_\xi(0)} \int_0^\infty R_\xi(\tau) d\tau = \frac{1}{\sigma_\xi^2} \int_0^\infty K_\xi(\tau) d\tau. \tag{1.28}$$

  This definition is particularly useful when $R_\xi(\tau)$ decays smoothly (e.g., Gaussian or exponential ACFs).

In this context, the *correlation interval* (or *correlation time*) quantifies the range of τ over which significant correlation persists.

Example: Consider a zero-mean process with Lorentzian ACF:

$$R_\xi(\tau) = \sigma_\xi^2 e^{-\alpha|\tau|}, \quad \alpha > 0 \tag{1.29}$$

Threshold method:

$$\tau_c = -\frac{1}{\alpha}\ln\beta, \quad \beta = 0.1 \;\Rightarrow\; \tau_c \approx 2.3/\alpha$$

Integral method:

$$\Delta\tau_e = \frac{1}{\sigma_\xi^2}\int_0^\infty \sigma_\xi^2 e^{-\alpha\tau}\, d\tau = \frac{1}{\alpha}.$$

The two definitions differ by a factor of ~ 2, but both scale as α^{-1}.

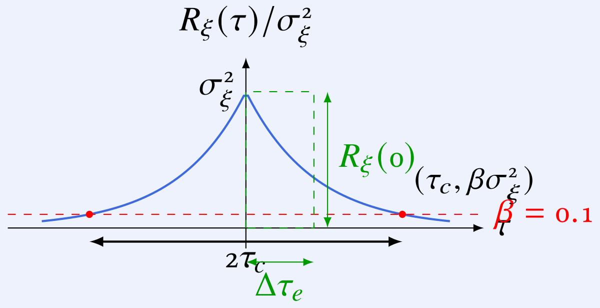


Figure: Correlation interval comparison.

Another aspect to consider, given two instantaneous signal values, is the relationship between their ACF and their mean-square difference. Let the *mean-square difference* (or *mean-square deviation*) between two signal values separated by lag τ be defined as

$$\varepsilon_\xi(\tau) = \mathbb{E}\left\{(\xi(t) - \xi(t+\tau))^2\right\}. \tag{1.30}$$

Expanding the squared term, we obtain

$$\varepsilon_\xi(\tau) = \mathbb{E}\left\{\xi^2(t)\right\} + \mathbb{E}\left\{\xi^2(t+\tau)\right\} - 2\mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\}. \tag{1.31}$$

For a *wide-sense stationary* process, the following properties hold:

- The second moment is time-invariant:

$$\mathbb{E}\left\{\xi^2(t)\right\} = \mathbb{E}\left\{\xi^2(t+\tau)\right\} = R_\xi(0).$$

- The autocorrelation function depends only on the lag:

$$\mathbb{E}\left\{\xi(t)\xi(t+\tau)\right\} = R_\xi(\tau).$$

Substituting these identities into (1.31) yields

$$\begin{aligned}\varepsilon_\xi(\tau) &= R_\xi(0) + R_\xi(0) - 2R_\xi(\tau)\\ &= 2R_\xi(0) - 2R_\xi(\tau).\end{aligned} \tag{1.32}$$

Rearranging (1.32), we obtain the fundamental identity

$$2R_\xi(\tau) + \varepsilon_\xi(\tau) = 2R_\xi(0). \tag{1.33}$$

Equation (1.33) shows that the autocorrelation function and the mean-square difference are linked by an exact affine relation:

$$\varepsilon_\xi(\tau) = 2\big(R_\xi(0) - R_\xi(\tau)\big).$$

Consequently:

- When $\tau = 0$,

  $$\varepsilon_\xi(0) = 0, \qquad R_\xi(0) = \mathbb{E}\left\{\xi^2(t)\right\}.$$

  Thus, $R_\xi(0)$ is the mean-square value, or average power, of the process.
- As the lag τ increases, if $R_\xi(\tau)$ decreases due to decorrelation, then $\varepsilon_\xi(\tau)$ increases according to

  $$\varepsilon_\xi(\tau) = 2\big(R_\xi(0) - R_\xi(\tau)\big).$$

  This behavior is typical for many physical random signals, but it is not a consequence of WSS alone.
- If the process is asymptotically uncorrelated, then

  $$\lim_{|\tau|\to\infty} K_\xi(\tau) = 0.$$

  Since

  $$R_\xi(\tau) = K_\xi(\tau) + \mu_\xi^2,$$

  it follows that

  $$\lim_{|\tau|\to\infty} R_\xi(\tau) = \mu_\xi^2.$$

  Moreover,

  $$R_\xi(0) = K_\xi(0) + \mu_\xi^2 = \sigma_\xi^2 + \mu_\xi^2.$$

  Consequently,

  $$\lim_{|\tau|\to\infty} \varepsilon_\xi(\tau) = 2R_\xi(0) - 2\mu_\xi^2 = 2\sigma_\xi^2.$$

In the zero-mean case, $\mu_\xi = 0$, this reduces to

$$\lim_{|\tau|\to\infty} \varepsilon_\xi(\tau) = 2R_\xi(0) = 2\sigma_\xi^2.$$

Therefore, for a real-valued WSS process, the ACF and the mean-square difference satisfy

$$\varepsilon_\xi(\tau) = 2\big(R_\xi(0) - R_\xi(\tau)\big).$$

Under asymptotic decorrelation, the mean-square difference approaches $2\sigma_\xi^2$, independently of the mean value.

Example: Consider the zero-mean random signal with Gaussian distribution, for which the ACF is a Dirac delta function scaled by the variance σ^2, i.e.,$R_\xi(\tau) = \sigma^2\delta(\tau)$.
The Mean Square Difference $\varepsilon_\xi(\tau)$, using the relationship, is computed as:

$$\varepsilon_\xi(\tau) = 2R_\xi(0) - 2R_\xi(\tau)$$

- At $\tau = 0$: $\quad \varepsilon_\xi(0) = 2\sigma^2 - 2\sigma^2 = 0$
  (No difference when comparing the signal to itself.)
- For $\tau \neq 0$: $\quad \varepsilon_\xi(\tau) = 2\sigma^2 - 2\cdot 0 = 2\sigma^2$
  (Since the signals are uncorrelated, the mean square difference is twice the variance.)

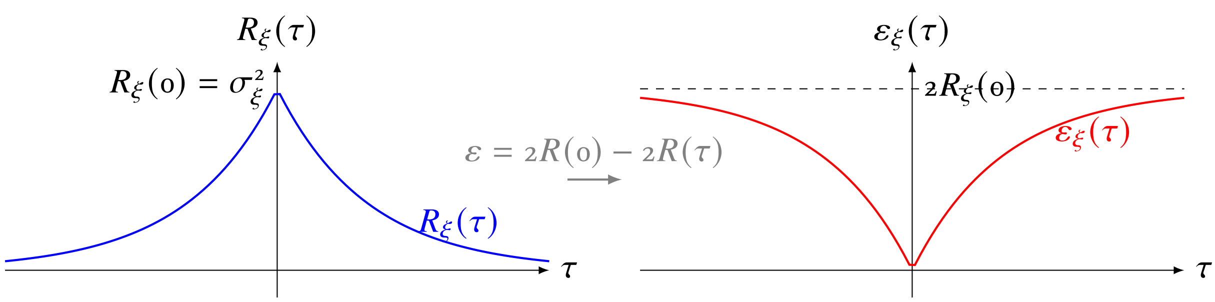


Figure 1.4: Complementary relationship between ACF and mean square difference for $R_\xi(\tau)=\sigma_\xi^2 e^{-\alpha|\tau|}$. As correlation decreases, mean square difference increases, maintaining $2R_\xi(\tau)+\varepsilon_\xi(\tau)=2R_\xi(0)$.

The mean square difference provides the analytical foundation for several applications:

- *Prediction theory:* The one-step-ahead prediction error variance equals $\varepsilon_\xi(\Delta t)$ for sampling interval Δt.
- *Interpolation:* Optimal interpolation (e.g., Wiener interpolation) minimizes $\varepsilon_\xi(\tau)$ between known samples.
- *Detection:* In matched filtering, the decision statistic involves $\varepsilon_\xi(\tau)$ between received and reference signals.
- *Stochastic differential equations:* For diffusion processes, $\mathbb{E}\left\{(X_{t+\Delta t} - X_t)^2\right\} \sim \Delta t$ as $\Delta t \to 0$ (quadratic variation).

Example: Find the ACF of the random signal: $\xi(t) = a\cos(\omega_0 t + \phi)$, $\phi \in [-\pi, \pi]$.
From the definition in Equation (1.18), it follows that

$$R_\xi(\tau) = \lim_{T\to\infty} \frac{1}{T} \int\limits_T a\cos(\omega_0 t + \phi)\, a\cos(\omega_0 t + \omega_0 \tau + \phi)\, dt$$

$$= \lim_{T\to\infty} \frac{a^2}{T} \int_T \frac{1}{2} \left(\cos(\omega_0 \tau + \phi) + \cos(2\omega_0 t + \omega_0 \tau + 2\phi)\right) dt$$

$$\text{where: } \frac{a^2}{2T} \int_T \cos(\omega_0 \tau + \phi)\, dt = \frac{a^2}{2} \cos\omega_0 \tau$$

$$\int_T \cos(2\omega_0 t + 2\phi + \omega_0 \tau)\, dt = \cos(2\phi + \omega_0 \tau) \int_T \cos 2\omega_0 t dt - \sin(2\phi + \omega_0 \tau) \int_T \sin 2\omega_0 dt$$

$$\text{Since it holds that: } \lim_{T\to\infty} \int_T \cos(k\omega_0 t)\, dt = \lim_{T\to\infty} \int_T \sin(k\omega_0 t)\, dt = 0, \quad \forall k \in \mathbb{Z}$$

$$\text{Then, } R_\xi(\tau) = \frac{a^2}{2} \cos\omega_0 \tau$$

Hypothetical patterns of the ACF

We consider several hypothetical shapes of the autocorrelation function $R_\xi(\tau)$. These shapes reveal second-order dependence properties of the corresponding real-valued WSS random signal $\xi(t)$. They should be interpreted as models of correlation structure, not as complete descriptions of the process.

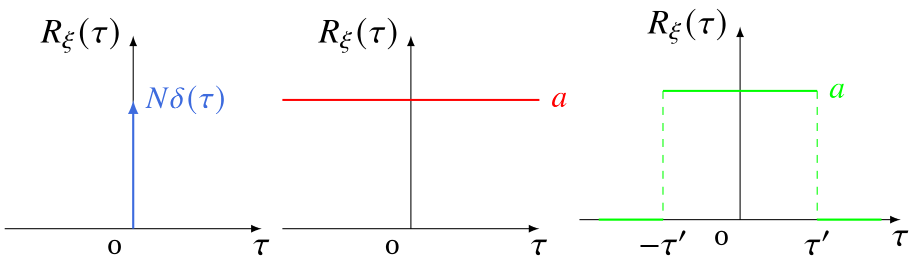


Figure 1.5: Hypothetical ACF patterns: full stochastic uncertainty (left), full stochastic certainty (middle), and a hybrid combination featuring intervals of certainty and uncertainty (right). Here a = const and N denotes the signal power.

- *Fully uncorrelated behavior.* Each instantaneous sample is correlated only with itself and uncorrelated with all other samples. The ACF is zero everywhere except at the origin (see left plot in Figure 1.5), given by:

$$R_\xi(\tau) \sim N\delta(\tau), \quad N \text{ is the signal power.}$$

- *Ergodicity with minimum uncertainty:* All signal samples are fully statistically dependent, resulting in a constant ACF across τ, as shown in the middle plot of Figure 1.5:

$$R_\xi(\tau) = const.$$

In this scenario, the process is no longer stochastic; the only remaining randomness is associated with the random variable scaling an otherwise deterministic signal $\xi(\cdot)$.

– *Pulse-shaped ACF idealization.* This scenario represents an intermediate form, characterized by complete correlation within a finite interval of duration $\tau' < \infty$, transitioning abruptly to zero correlation thereafter as $\tau \longrightarrow \infty$:

$$R_\xi(\tau) = \begin{cases} const, & |\tau| \le \tau' \\ 0, & \text{otherwise.} \end{cases} \tag{1.34}$$

The signal shown in the right plot of Figure 1.5 represents an abstract model, but it is not a valid ACF of a WSS process because it violates the non-negative-definiteness condition in Equation (1.24). Indeed, its Fourier transform is

$$S(\omega) = 2\int_0^{\tau'} a\cos(\omega\tau)\,d\tau = \frac{2a}{\omega}\sin(\omega\tau') \not\ge 0, \quad \forall\,\omega.$$

This function changes sign as ω varies, and therefore cannot be a valid PSD. By Bochner's theorem, any valid ACF must have a non-negative spectral measure. Hence, the rectangular ACF serves only as a conceptual illustration of what fails when non-negative definiteness is ignored.

Finite-support (truncated) ACF. In contrast to the inadmissible rectangular idealization in Equation (1.34), many practical stationary models produce ACFs that vanish beyond a finite lag without being constant inside the support. A canonical example is the first-order moving-average process, where the current value is represented as a linear combination of present and past white-noise components:

$$x_t = \mu + \epsilon_t + \beta\epsilon_{t-1}, \tag{1.35}$$

where μ denotes the mean (often set to zero for simplicity), $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$ represents zero-mean Gaussian noise with variance σ^2 (such that $\mathbb{E}\{\epsilon_t\epsilon_{t-k}\} = \sigma^2$ for $k = 0$ and 0 otherwise), and β is the model parameter.

The ACF at lag τ is:

$$\begin{aligned}
R_x(\tau) &= K_x(\tau)/K_x(0) \\
\text{where } K_x(0) &= \mathbb{E}\{x_t x_t\} = \mathbb{E}\{(\epsilon_t + \beta\epsilon_{t-1})^2\} = \mathbb{E}\left\{\epsilon_t^2 + 2\beta\epsilon_{t-1}\epsilon_t + \beta^2\epsilon_{t-1}^2\right\} \\
&= \mathbb{E}\left\{\epsilon_t^2\right\} + 2\mathbb{E}\{\beta\epsilon_{t-1}\epsilon_t\} + \beta^2\mathbb{E}\left\{\epsilon_{t-1}^2\right\} \\
K_x(0) &= \sigma^2 + \beta^2\sigma^2 = \sigma^2(1+\beta^2) \\
\text{at lag } \tau = 1,\ K_x(1) &= \mathbb{E}\{x_t x_{t+1}\} = \mathbb{E}\{(\epsilon_t + \beta\epsilon_{t-1})(\epsilon_{t+1} + \beta\epsilon_t)\} \\
&= \mathbb{E}\{\epsilon_t\epsilon_{t+1}\} + \beta\mathbb{E}\left\{\epsilon_t^2\right\} + \beta\mathbb{E}\{\epsilon_{t-1}\epsilon_{t+1}\} + \beta^2\mathbb{E}\{\epsilon_t\epsilon_{t-1}\} \\
&= \beta\mathbb{E}\left\{\epsilon_t^2\right\} + 0 + 0 + 0 = \beta\sigma^2.
\end{aligned}$$

At lag $\tau \geq 2$, the following values for ACF are obtained:

$$K_x(\tau) = \mathbb{E}\left\{(\epsilon_t + \beta\epsilon_{t-1})(\epsilon_{t+\tau} + \beta\epsilon_{t+\tau-1})\right\}$$
$$= \mathbb{E}\left\{\epsilon_t\epsilon_{t+\tau}\right\} + \beta\mathbb{E}\left\{\epsilon_t\epsilon_{t+\tau-1}\right\} + \beta\mathbb{E}\left\{\epsilon_{t-1}\epsilon_{t+\tau}\right\} + \beta^2\mathbb{E}\left\{\epsilon_{t-1}\epsilon_{t+\tau-1}\right\} = 0,\ |\tau| \geq 2,$$

$$\text{then, } R(\tau) = \begin{cases} 1, & \tau = 0, \\ \dfrac{\beta}{(1+\beta^2)}, & |\tau| = 1, \\ 0, & |\tau| > 1. \end{cases}$$

Consequently, the ACF is nonzero only up to lag 1, reflecting the model's dependence on just one preceding noise term. Crucially, this *truncated* shape is admissible because its Fourier transform, the PSD

$$S_x(\omega) = \sigma^2\left|1 + \beta e^{-j\omega}\right|^2 = \sigma^2\left(1 + \beta^2 + 2\beta\cos\omega\right), \tag{1.36}$$

is manifestly non-negative for every real β and all ω. (The constraint $|\beta| < 1$ is required only for filter invertibility, not for PSD validity.) This highlights the methodological distinction between a *rectangular* ACF (which violates non-negative definiteness) and a *finite-lag* or *truncated* ACF (which can be valid if the resulting PSD remains non-negative).

Another common pattern of truncated ACF is one in which statistical dependence weakens monotonically as the time lag increases: $R_\xi(\tau) = f(\tau)$, where $f(\cdot)$ is a decreasing function of τ. A paradigmatic example is the first-order autoregressive (AR(1)) process

$$x_t = \alpha x_{t-1} + \epsilon_t, \quad t \in \mathbb{Z}, \tag{1.37}$$

where x_t denotes the signal at time t, $\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$ is Gaussian noise, and $\alpha \in \mathbb{R}$ is the autoregressive parameter. Depending on the value of α, the following cases arise:

- *Stationary regime ($|\alpha| < 1$):* The process is wide-sense stationary. Assuming zero mean, the covariance satisfies the recursion

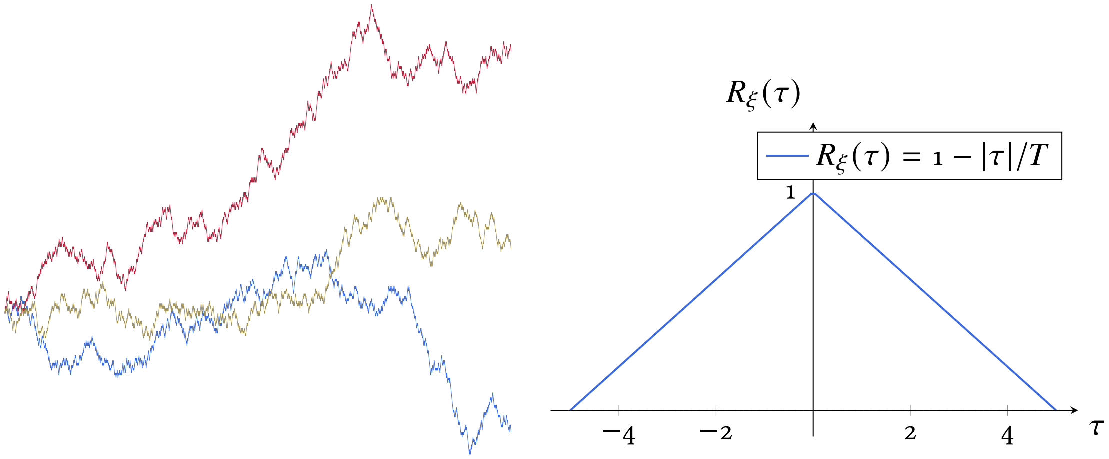
  $$K_x(k) = \mathbb{E}\left\{x_t x_{t-k}\right\} = \alpha\mathbb{E}\left\{x_{t-1}x_{t-k}\right\} + \mathbb{E}\left\{\epsilon_t x_{t-k}\right\} = \alpha K_x(k-1),$$

  since $\mathbb{E}\left\{\epsilon_t x_{t-k}\right\} = 0$ for $k \geq 1$. Solving iteratively with $K_x(0) = \sigma_\epsilon^2/(1-\alpha^2)$ yields

  $$R_x(k) = \frac{K_x(k)}{K_x(0)} = \alpha^{|k|}, \qquad k \in \mathbb{Z}. \tag{1.38}$$

  The term α acts as a fading factor that determines the strength of dependence on previous values; the ACF decays exponentially and is never exactly zero for finite lag.

- *Non-stationary regime ($|\alpha| = 1$):* For $\alpha = 1$, Equation (1.37) becomes a random walk. Because the variance grows linearly with time ($\mathbb{E}\left\{x_t^2\right\} = t\sigma_\epsilon^2$), the process is *not* wide-sense stationary and does not possess a time-invariant ACF $R(\tau)$. The triangular pulse sometimes associated with this case actually belongs to a *finite moving-average* (e.g., $x_t = \sum_{i=0}^{T-1} \epsilon_{t-i}$), which is stationary. The random walk itself serves as a boundary case that separates stationary AR(1) processes from explosive ones ($|\alpha| > 1$).



a) Simulated random paths b) Approximated ACF

Figure 1.6: Finite moving-average process and its triangular normalized ACF. This triangular ACF should not be interpreted as the ACF of a random walk, because a random walk is not wide-sense stationary. RandomWalkAFC

Example: Find the ACF of the ergodic square pulse train shown in Figure 1.8 (top row).
For a unit-amplitude unipolar rectangular pulse train of period $T = 2\pi/\omega_0$ and pulse width Δt (duty cycle $D = \Delta t/T$), the even Fourier series is

$$\xi(t) = \frac{\omega_0\,\Delta t}{2\pi} + \frac{2}{\pi}\sum_{n=1}^{\infty}\frac{1}{n}\sin\left(n\,\frac{\omega_0\,\Delta t}{2}\right)\cos(n\,\omega_0\,t).$$

For a deterministic periodic signal the time-averaged ACF can be obtained directly from the Fourier coefficients. Because the signal is even, its shifted copy $\xi(t+\tau)$ has the identical cosine coefficients ($a_\xi = a_\eta$), and the ACF reduces to the autocorrelation of the Fourier series:

$$R_\xi(\tau) = \left(\frac{a_0}{2}\right)^2 + \frac{1}{2}\sum_{n=1}^{\infty} a_n^2\cos(n\omega_0\tau). \tag{1.39}$$

Substituting $a_0 = \omega_0\Delta t/\pi$ and $a_n = \dfrac{2}{n\pi}\sin\left(n\dfrac{\omega_0\Delta t}{2}\right)$ gives

$$\begin{aligned} R_\xi(\tau) &= \left(\frac{\omega_0\Delta t}{2\pi}\right)^2 + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\sin^2\left(n\frac{\omega_0\Delta t}{2}\right)\cos(n\omega_0\tau) \\ &= D^2 + \frac{2}{\pi^2}\sum_{n=1}^{\infty}\frac{1}{n^2}\sin^2\left(n\frac{\omega_0\Delta t}{2}\right)\cos(n\omega_0\tau). \end{aligned}$$

To evaluate the series in closed form we use the identity [1]

$$\sum_{n=1}^{\infty}\frac{\sin^2(n\alpha)}{n^2}\cos(nx) = \begin{cases} -\dfrac{\pi}{4}x + \dfrac{\pi\alpha}{2} - \dfrac{\alpha^2}{2}, & 0 < x < 2\alpha, \\ -\dfrac{\alpha^2}{2}, & 2\alpha < x < \pi, \end{cases}$$

342 with $\alpha = \omega_0\Delta t/2$ and $x = \omega_0\tau$, and extend the result by evenness ($R_\xi(-\tau) = R_\xi(\tau)$) and

periodicity ($R_\xi(\tau+T) = R_\xi(\tau)$). This yields the piecewise closed form

$$R_\xi(\tau) = \begin{cases} \dfrac{\Delta t - |\tau|}{T}, & |\tau| \leq \Delta t, \\ 0, & \Delta t < |\tau| \leq T/2, \end{cases} \qquad \text{(extended periodically with period } T\text{).} \tag{1.40}$$

PyTask: Generate stationary signals with ACF, given by the Gaussian shape:

$$R_\xi(\tau)=\sigma^2 e^{-\alpha\tau^2}, \tag{1.41}$$

where σ^2 is the process variance and $\alpha>0$ is a parameter controlling the decay rate of ACF (see Figure 1.7).

343

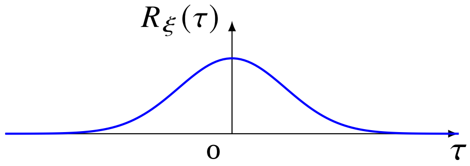


Figure 1.7: Gaussian-shaped ACF $R_\xi(\tau) = \sigma^2 e^{-\alpha\tau^2}$ (illustrative).

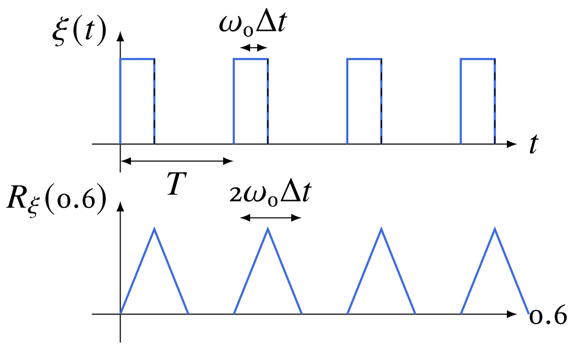


Figure 1.8: Square pulse periodic train (top row) and its ACF (bottom row). `01a4bCorrParFunction`

Pytask: PyTask

Comparison between different algorithms for ACF estimation

Show that ACF for

$$\xi(t) = \cos(\omega_0 t + \phi) + \eta(t),$$

where $\eta \sim \mathcal{N}_\eta(0, \sigma_\eta^2)$, zero-mean perturbation ruled by Gaussian PDF, the ACF is:

$$R_\xi\,(\tau) = \cos\omega\tau + \sigma_\eta^2\delta(\tau)/2$$

1. Experimental Approaches for comparing estimating methods of ACF.
   - ↪ Methods for Autocorrelation Function (ACF) Estimation
   - ↪ Generate a signal composed of three harmonics plus a zero mean random signal with 1024 samples
   - ↪ Test the stationarity conditions and linear dependence for the last two items
   - ↪ Integrate a timer

344

- ↪ [NB] define the sampling frequency and use it to calculate the time step and adjust the time vector. Fix the following values: Number of samples: $N = 1024$, Frequencies: $[5, 10, 25]$ Hz, Amplitudes: $[1.0, 0.5, 0.3]$, Sampling frequency: 1024 Hz, Noise amplitude: 0.2

  — (Note that the time step is fixed as: Δt=1/sampling frequency) —

2. Sample Autocorrelation
   - ↪ Sample Autocorrelation (Direct Method) method mathematical description in latex
   - ↪ Integrate Sample Autocorrelation (Direct Method)
   - ↪ provide an example in python
   - ↪ Mitigate the bias that can arise due to the limited estimation window
   - ↪ [NB] plot signal without noise

     — fix manually minor issues —–
   - ↪ [NB] Optimize the Python code for efficiency
   - ↪ [NB] convert to pytorch fully

     — explain differences between theoretical and experimental ACF results —

     01a1CorrFunction,01a3CorrFunction
3. Alternative Estimation Methods
   - ✓ Fourier Transform Method (via Power Spectral Density)
   - ✓ Yule-Walker Equations
   - ✓ Periodogram-Based Estimation
   - ✓ Maximum Likelihood Estimation (MLE)
   - ✓ Robust Estimation Techniques
4. Integration and Visualization
   - ↪ Integrate into a single notebook all validated estimation methods
   - ↪ Plot together all estimated ACF for comparison
5. Estimate ACF for a multivariate random process. 01a2CorrFunction
   - ↪ Plot together all estimated ACF for comparison
   - ↪ Measure the difference between the estimated and the theoretical results.

1.3 Power Spectral Density

Windowed mean-square value

A wide-sense stationary random signal $\xi(t)$ generally cannot be represented directly by the ordinary Fourier transform

$$\Xi(\omega) = \int_{-\infty}^{\infty} \xi(t) e^{-j\omega t}\, dt. \tag{1.42}$$

A stationary process typically has finite average power but infinite total energy; therefore, its sample paths are generally not absolutely integrable over $\mathbb{R}$, and Equation (1.42) is not well defined as an ordinary Fourier transform.

To obtain a meaningful spectral representation, observe the process is observed over a finite interval window, as defined in Equation (1.4a):

$$\xi_T(t) = \mathrm{rect}_T(t)\xi(t). \tag{1.43a}$$

$$\text{with FT equals to} \quad \Xi_T(\omega) \int_{-T}^{T} \xi(t) e^{-j\omega t}\, dt. \tag{1.43b}$$

For a second-order process,

$$\mathbb{E}\left\{\int_{-T}^{T} \xi^2(t)\, dt\right\} = \int_{-T}^{T} \mathbb{E}\left\{\xi^2(t)\right\}\, dt < \infty,$$

so the windowed observation has finite mean-square energy over every finite interval.

Applying Parseval's theorem to $\xi_T(t)$ gives

$$\int_{-T}^{T} \xi^2(t)\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\Xi_T(\omega)|^2\, d\omega. \tag{1.44}$$

Dividing by the observation length $2T$,

$$\frac{1}{2T} \int_{-T}^{T} \xi^2(t)\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{|\Xi_T(\omega)|^2}{2T}\, d\omega. \tag{1.45}$$

The left-hand side is the time-averaged mean-square value over $[-T, T]$:

$$\overline{\{\xi^2\}}_T = \frac{1}{2T} \int_{-T}^{T} \xi^2(t)\, dt.$$

Taking ensemble expectations in Equation (1.45) yields

$$\mathbb{E}\left\{\frac{1}{2T} \int_{-T}^{T} \xi^2(t)\, dt\right\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{\mathbb{E}\left\{|\Xi_T(\omega)|^2\right\}}{2T}\, d\omega. \tag{1.46}$$

For a WSS process, $\mathbb{E}\left\{\xi^2(t)\right\} = R_\xi(0)$, and therefore

$$\mathbb{E}\left\{\frac{1}{2T} \int_{-T}^{T} \xi^2(t)\, dt\right\} = R_\xi(0).$$

This motivates the power spectral density

$$S_\xi(\omega) = \lim_{T\to\infty} \frac{\mathbb{E}\left\{|\Xi_T(\omega)|^2\right\}}{2T}, \tag{1.47}$$

whenever the limit exists in the appropriate sense. Consequently,

$$R_\xi(0) = \mathbb{E}\left\{\xi^2(t)\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_\xi(\omega)\, d\omega. \tag{1.48}$$

For a zero-mean WSS process, $R_\xi(0) = \sigma_\xi^2$. For a nonzero-mean process,

$$R_\xi(0) = \sigma_\xi^2 + \mu_\xi^2,$$

so the PSD includes the contribution of the constant component at zero frequency.

Under suitable ergodicity assumptions, the ensemble-averaged PSD can be estimated from a single sufficiently long realization by the finite-window periodogram

$$\widehat{S}_{\xi,T}(\omega) = \frac{|\Xi_T(\omega)|^2}{2T}. \tag{1.49a}$$

$$\text{Thus,}\quad \overline{\{\xi^2\}}_T = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{S}_{\xi,T}(\omega)\, d\omega. \tag{1.49b}$$

This last identity is a finite-window energy balance for one realization. The replacement of ensemble averages by time averages requires ergodicity in the corresponding second-order quantities, not merely ergodicity in the mean.

The *Power Spectral Density (PSD)* $S_\xi(\omega)$ is defined as the asymptotic (ensemble-averaged in Equation (1.47)) finite-time periodogram, i.e., the squared magnitude of the truncated FT normalized by the observation length:

$$S_\xi(\omega) \triangleq \lim_{T\to\infty} \frac{\mathbb{E}\left\{|\Xi_T(\omega)|^2\right\}}{2T}, \tag{1.50}$$

whenever the limit exists in the appropriate sense.

Therefore, the PSD is not the mean-square value itself; rather, it is the spectral density whose integral gives the mean-square value of the process:

$$R_\xi(0) = \mathbb{E}\left\{\xi^2(t)\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_\xi(\omega)\, d\omega. \tag{1.51}$$

For a zero-mean WSS process, $R_\xi(0) = \sigma_\xi^2$, and therefore

$$\sigma_\xi^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_\xi(\omega)\, d\omega. \tag{1.52}$$

For a nonzero-mean WSS process,

$$R_\xi(0) = \sigma_\xi^2 + \mu_\xi^2.$$

In that case, the constant component contributes a spectral line at $\omega = 0$. Under the angular-frequency convention used here, this contribution is $2\pi\mu_\xi^2\delta(\omega)$. Thus, if $S_\xi(\omega)$ denotes the full PSD including this spectral line, then Equation (1.51) gives the total mean-square value. If the DC component is removed, then the remaining PSD integrates to the variance:

$$\sigma_\xi^2 = \frac{1}{2\pi}\int_{-\infty}^{\infty} S_{\xi-\mu_\xi}(\omega)\, d\omega.$$

Remark. If the signal $\xi(t)$ is measured in $[V]$, then, the PSD $S_\xi(\omega)$ has units $[V^2/(\text{rad/s})]$. Likewise, if ordinary frequency $f = \omega/(2\pi)$ is used, the PSD $S_\xi(f)$ has units $[V^2/\text{Hz}]$. So, the two PSD conventions are related by

$$S_\xi(f) = S_\xi(\omega)\, d\omega/df = 2\pi S_\xi(\omega), \qquad \omega = 2\pi f.$$

For a real-valued WSS process, the PSD has the following properties:

(a) $S_\xi(\omega) \in \mathbb{R}$: it is real-valued whenever it exists as an ordinary function.

(b) $S_\xi(\omega) \geq 0$: it is non-negative. In general, the PSD need not be bounded and may contain singular components, such as impulses.

(c) $S_\xi(\omega) = S_\xi(-\omega)$: it is even for real-valued WSS processes. Therefore, when a PSD is modeled by an even rational function, it may be written as a rational function of ω^2[1]:

$$S_\xi(\omega) = \frac{s_0\left(\omega^{2n} + a_{2n-2}\omega^{2n-2} + \cdots + a_2\omega^2 + a_0\right)}{\omega^{2m} + b_{2m-2}\omega^{2m-2} + \cdots + b_2\omega^2 + b_0}. \tag{1.53}$$

Since the PSD characterizes the distribution of average power over frequency, one may define normalized spectral moments when the required integrals are finite. For a zero-mean real-valued WSS process with even PSD,

$$\sigma_\xi^2 = \frac{1}{\pi}\int_0^{\infty} S_\xi(\omega)\, d\omega.$$

Thus, a normalized one-sided spectral weighting function can be defined as

$$p_\omega(\omega) = \frac{S_\xi(\omega)}{\pi\sigma_\xi^2}, \qquad \omega \geq 0, \quad \text{which satisfies} \int_0^{\infty} p_\omega(\omega)\, d\omega = 1.$$

This one-sided weighting function describes how the average power is distributed over nonnegative angular frequencies. The first spectral moment and the spectral spread are then

$$m_{1\omega} = \int_0^{\infty} \omega\, p_\omega(\omega)\, d\omega = \frac{1}{\pi\sigma_\xi^2}\int_0^{\infty} \omega S_\xi(\omega)\, d\omega, \tag{1.54a}$$

$$\sigma_\omega^2 = \int_0^{\infty} (\omega - m_{1\omega})^2 p_\omega(\omega)\, d\omega = \frac{1}{\pi\sigma_\xi^2}\int_0^{\infty} (\omega - m_{1\omega})^2 S_\xi(\omega)\, d\omega. \tag{1.54b}$$

Here, $m_{1\omega}$ has units of rad/s, while σ_ω^2 has units of $(\text{rad/s})^2$. The quantity σ_ω^2 measures the spread of spectral power around the mean angular frequency $m_{1\omega}$.

[1] This is a modeling form for even rational spectra, not a general requirement for all valid PSDs.

The *effective angular bandwidth* may be defined as

$$\Delta\omega_e = \frac{1}{S_\xi(0)} \int_{-\infty}^{\infty} S_\xi(\omega)\, d\omega,$$

provided that $S_\xi(0)$ is finite and strictly positive, and that the integral is finite. Using Equation (1.51), this becomes

$$\Delta\omega_e = \frac{2\pi R_\xi(0)}{S_\xi(0)}.$$

If the correlation interval is defined by

$$\Delta\tau = \frac{1}{R_\xi(0)} \int_0^{\infty} R_\xi(\tau)\, d\tau,$$

and $R_\xi(\tau)$ is even and integrable, then

$$S_\xi(0) = \int_{-\infty}^{\infty} R_\xi(\tau)\, d\tau = 2\int_0^{\infty} R_\xi(\tau)\, d\tau.$$

Therefore,

$$\Delta\tau = \frac{S_\xi(0)}{2R_\xi(0)}.$$

Consequently, with the above definitions and the angular-frequency convention,

$$\Delta\omega_e\, \Delta\tau = \pi.$$

Relationship between $S_\xi(\omega)$ and $R_\xi(\tau)$

The autocorrelation function describes second-order dependence in the time domain, whereas the PSD describes the distribution of average power in the frequency domain. For a WSS process, these two descriptions form a Fourier transform pair.

Consider the finite-window Fourier transform

$$\Xi_T(\omega) = \int_{-T}^{T} \xi(t) e^{-j\omega t}\, dt.$$

For a real-valued process, $\Xi_T^*(\omega) = \Xi_T(-\omega)$, and the PSD may be defined as

$$S_\xi(\omega) = \lim_{T\to\infty} \frac{\mathbb{E}\left\{|\Xi_T(\omega)|^2\right\}}{2T}. \tag{1.55}$$

Expanding the squared magnitude gives

$$\begin{aligned} \mathbb{E}\left\{|\Xi_T(\omega)|^2\right\} &= \mathbb{E}\left\{\left(\int_{-T}^{T} \xi(t_1) e^{-j\omega t_1}\, dt_1\right)\left(\int_{-T}^{T} \xi(t_2) e^{j\omega t_2}\, dt_2\right)\right\} \\ &= \int_{-T}^{T}\int_{-T}^{T} \mathbb{E}\left\{\xi(t_1)\xi(t_2)\right\} e^{-j\omega(t_1-t_2)}\, dt_1 dt_2. \end{aligned} \tag{1.56}$$

For a real-valued WSS process,

$$\mathbb{E}\left\{\xi(t_1)\xi(t_2)\right\} = R_\xi(t_1 - t_2) = R_\xi(t_2 - t_1).$$

Then the double integral over the square $[-T, T]\times[-T, T]$ can be rewritten, making $\tau = t_1 - t_2$, as

$$\frac{\mathbb{E}\left\{|\Xi_T(\omega)|^2\right\}}{2T} = \int_{-2T}^{2T} \left(1 - \frac{|\tau|}{2T}\right) R_\xi(\tau) e^{-j\omega\tau}\, d\tau. \tag{1.57}$$

The factor $1 - |\tau|/2T$ is the finite-window triangular weighting induced by the overlap of the two integration intervals. Taking $T \to \infty$, and under the usual conditions that justify the limiting operation, this factor tends to one for each fixed τ.

Hence, the following relationship between the time and frequency domains holds:

$$S_\xi(\omega) = \int_{-\infty}^{\infty} R_\xi(\tau) e^{-j\omega\tau}\, d\tau \tag{1.58a}$$

417 Conversely, $$R_\xi(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_\xi(\omega) e^{j\omega\tau}\, d\omega. \tag{1.58b}$$

The Fourier-transform pair in Equations (1.58a) and (1.58b) is the *Wiener–Khinchin theorem*, as illustrated in Figure 1.9).

For two jointly WSS real-valued processes $\xi(t)$ and $\eta(t)$, the Wiener–Khinchin pair is defined as:

$$R_{\xi\eta}(\tau) = \mathbb{E}\left\{\xi(t)\eta(t+\tau)\right\}$$

$$R_{\xi\eta}(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\xi\eta}(\omega) e^{j\omega\tau}\, d\omega. \tag{1.59a}$$

$$S_{\xi\eta}(\omega) = \int_{-\infty}^{\infty} R_{\xi\eta}(\tau) e^{-j\omega\tau}\, d\tau, \tag{1.59b}$$

Remark. If $R_{\xi\eta}(\tau) = 0,\ \forall\tau$, then $S_{\xi\eta}(\omega) = 0$. This is stronger than uncorrelatedness in the covariance sense unless the processes are zero-mean. For zero-mean processes, cross-correlation and cross-covariance coincide.

For the autocorrelation of a real-valued WSS process, $R_\xi(\tau)$ is even, so the Wiener–Khinchin pair reduces to the Fourier cosine transform:

$$S_\xi(\omega) = 2\int_0^{\infty} R_\xi(\tau)\cos(\omega\tau)\, d\tau, \tag{1.60a}$$

$$R_\xi(\tau) = \frac{1}{\pi}\int_0^{\infty} S_\xi(\omega)\cos(\omega\tau)\, d\omega. \tag{1.60b}$$

For complex-valued processes, the full complex-exponential form in Equations (1.58a) and (1.58b) must be retained, with conjugation applied to the second signal in the correlation definition.

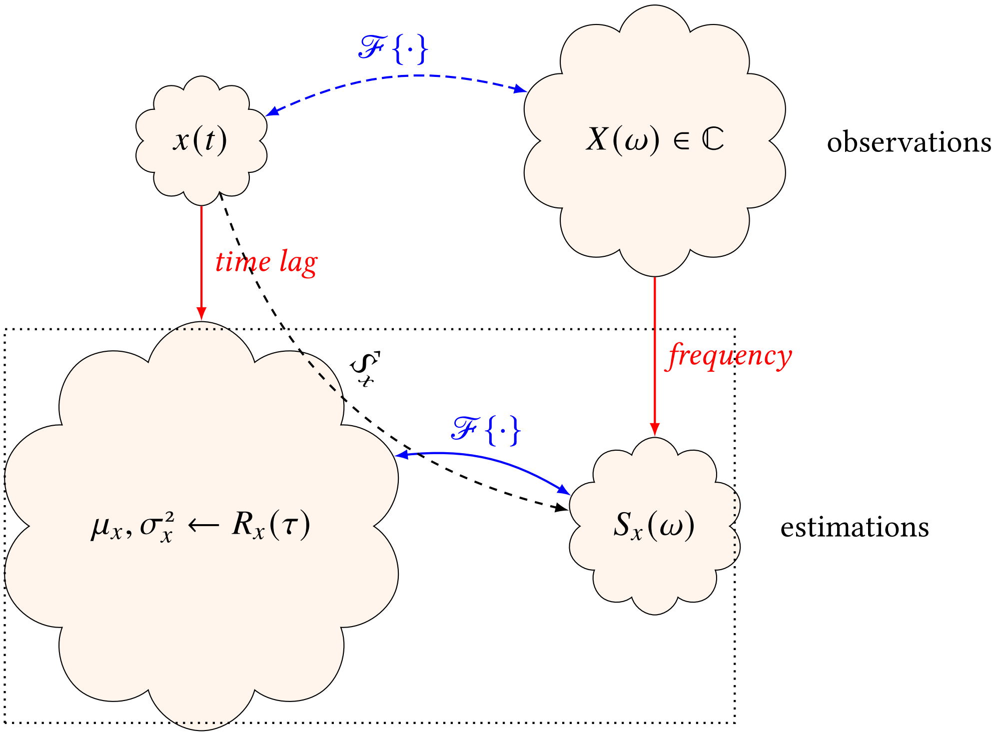


Figure 1.9: Relationship between ACF and PSD: $\tau \mapsto \omega$

Hypothetical patterns of spectra

The Wiener–Khinchin framework relates spectral shapes to ACF shapes, allowing one to interpret second-order dependence either in the lag domain or in the frequency domain. Under the angular-frequency convention

$$S_\xi(\omega) = \int_{-\infty}^{\infty} R_\xi(\tau) e^{-j\omega\tau}\, d\tau, \qquad R_\xi(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_\xi(\omega) e^{j\omega\tau}\, d\omega,$$

two limiting idealizations are especially useful:

– *Flat spectrum.* If

$$\begin{aligned} & S_\xi(\omega) = N, \qquad -\infty < \omega < \infty, \\ \text{then,}\quad & R_\xi(\tau) = N\delta(\tau). \end{aligned}$$

This the process has equal spectral density at all angular frequencies and is uncorrelated at nonzero lags.

– *Spectral line at zero frequency.* If

$$\begin{aligned} & S_\xi(\omega) = 2\pi N\, \delta(\omega), \\ \text{then,}\quad & R_\xi(\tau) = N. \end{aligned}$$

This corresponds to a constant ACF, meaning that all time instants remain fully correlated. A simple model is $\xi(t) = C$, where C is a random variable independent of t. In that case, $N = \mathbb{E}\{C^2\}$.

Example: Revisiting the stationary process with exponential ACF ??, whose corresponding PSD is Lorentzian, the Wiener–Khinchin theorem gives

Using the Wiener–Khinchin relation, the PSD is obtained via the Fourier cosine transform:

$$S_\xi(\omega) = 2\int_0^\infty \sigma_\xi^2 e^{-\alpha\tau}\cos(\omega\tau)\,d\tau$$
$$= 2\sigma_\xi^2 \frac{\alpha}{\alpha^2+\omega^2}.$$

Asymptotic behavior:

$$\lim_{\alpha\to 0^+} S_\xi(\omega) = \lim_{\alpha\to 0^+} \frac{2\sigma_\xi^2\,\alpha}{\alpha^2+\omega^2} = 0, \quad \omega \neq 0,$$

while at $\omega = 0$,

$$S_\xi(0) = \frac{2\sigma_\xi^2}{\alpha} \xrightarrow[\alpha\to 0^+]{} \infty.$$

The limit converges in the distributional sense to a Dirac delta:

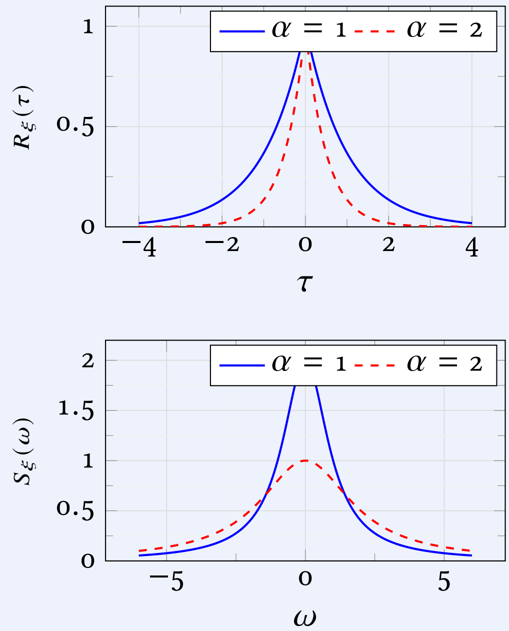


Figure 1.10: Autocorrelation and PSD.

$$\lim_{\alpha\to 0^+} S_\xi(\omega) = 2\pi\sigma_\xi^2\,\delta(\omega).$$

The 3 dB bandwidth is $\Delta\omega = \alpha$, and the total power integrates to σ_ξ^2, recovering (1.52).

439

440 **White Gaussian Noise.**

White Gaussian Noise (WGN): This model describes an idealized ergodic signal with a constant PSD over the entire frequency range, as shown in Figure 1.11(b):

$$S_\xi(\omega) = N_0/2, \quad \omega \in \mathbb{R}, \tag{1.61}$$

where $N_0 > 0$ is a constant with units of power per unit bandwidth (e.g., [W/Hz] or [V^2/Hz]). The factor 1/2 in (1.61) arises from the *two-sided* PSD convention, which accounts for both positive and negative frequencies. Still, the *one-sided* PSD is also defined:

$$S_\xi^+(\omega) = N_0, \quad \omega \geq 0, \tag{1.62}$$

441

Either model, (1.61) or (1.62), is known as *white noise* due to its spectral resemblance to white light, which contains all visible frequencies with equal intensity.
Throughout this text, we use the two-sided convention unless otherwise stated, for which the ACF follows from Equation (1.58b) as illustrated in Figure 1.11(a):

$$R_\xi(\tau) = \mathscr{F}^{-1}\left\{S_\xi(\omega)\right\} = \frac{1}{2\pi}\int_{-\infty}^{\infty}\frac{N_0}{2}e^{j\omega\tau}d\omega = \frac{N_0}{4\pi}\cdot 2\pi\delta(\tau)$$

$$\text{then, } R_\xi(\tau) = \frac{N_0}{2}\delta(\tau), \tag{1.63}$$

442 From the model Equation (1.63), it follows the following properties of WGN:

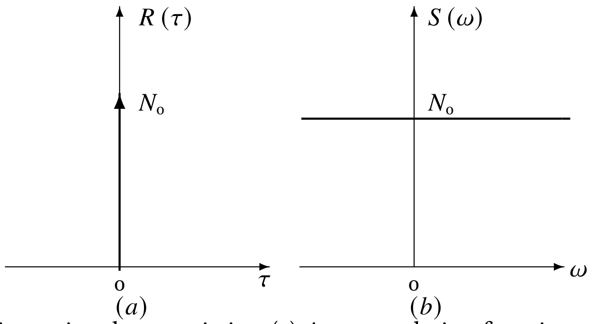


Figure 1.11: White Gaussian noise characteristics: (a) Autocorrelation function as a Dirac delta, and (b) constant power spectral density across all frequencies.

(i) *Uncorrelated samples:*
From (1.63), $R_\xi(\tau)=0,\ \forall\tau\neq 0$, meaning any two distinct time samples are uncorrelated:

$$\mathbb{E}\left\{\xi(t_1)\xi(t_2)\right\} = 0, \quad t_1 \neq t_2. \tag{1.64}$$

For Gaussian processes, uncorrelatedness implies *statistical independence*, so WGN samples at different times are totally independent random variables.

(ii) *Infinite variance:*
Evaluating the ACF at $\tau=0$ gives the signal power:

$$R_\xi(0) = \mathbb{E}\left\{\xi^2(t)\right\} = \frac{N_0}{2}\delta(0) \rightarrow \infty. \tag{1.65}$$

This infinite variance is a mathematical idealization reflecting the fact that WGN contains *infinite power.*

(iii) *Infinite bandwidth:* The PSD is non-zero for all ω, meaning the signal has infinite bandwidth.

(iv) *Flat PSD:* The constant PSD implies that each frequency component contributes equally to the signal's power and therefore, holds similar uncertainty.

Consequently, true white noise is *not physically realizable* because: i) Infinite variance implies infinite energy per sample, ii) Infinite bandwidth exceeds any physical system's capacity, and iii) All physical measurement devices have finite bandwidth.

1.4 Finite-Band Random Signals

Band-limited white noise

Since all signal processing devices operate within a finite bandwidth $\Delta\omega$, any noise model at their output is also bandwidth-limited. Assuming a constant PSD over $|\omega| < \Delta\omega$, the resulting model is

$$S(\omega) = \begin{cases} N_0/2, & |\omega| < \Delta\omega \\ 0, & \text{otherwise,} \end{cases} \tag{1.66}$$

whose ACF follows from the inverse Fourier transform:

$$R(\tau) = \frac{1}{2\pi}\int_{-\Delta\omega}^{\Delta\omega} \frac{N_0}{2} e^{j\omega\tau}\, d\omega = \frac{N_0\Delta\omega}{2\pi}\,\mathrm{sinc}\left(\frac{\Delta\omega\,\tau}{\pi}\right), \tag{1.67}$$

where we adopt the normalized sinc function $\mathrm{sinc}(x) \triangleq \sin(\pi x)/(\pi x)$.

Because its bandwidth is narrower than that of ideal WGN, this model is called *finite-bandwidth white noise*. Its properties are:

- *Finite output power*: The total power is

$$P_\xi = R_\xi(0) = \frac{1}{2\pi}\int_{-\Delta\omega}^{\Delta\omega} \frac{N_0}{2}\, d\omega = \frac{N_0\Delta\omega}{2\pi}. \tag{1.68}$$

- *Correlated instantaneous values:* Unlike ideal WGN, samples separated by $\tau < \pi/\Delta\omega$ are correlated. The first zero crossing of the ACF occurs at $\tau_0 = \pi/\Delta\omega$ (where the sinc argument equals 1), defining the *correlation time*.
- *Limiting behavior*: As $\Delta\omega \to \infty$, $\mathrm{sinc}(\Delta\omega\tau/\pi) \to \delta(\tau)$ in the distributional sense, recovering ideal WGN.

Inadmissibility of a rectangular ACF. There does not exist a stationary process with a piecewise constant ACF of finite support:

$$R_\xi(\tau) = \begin{cases} N_0, & |\tau| \le \tau' \\ 0, & \text{otherwise.} \end{cases} \tag{1.69}$$

The necessary conditions for a valid ACF are checked as follows:

$$\begin{aligned} &\textit{Even symmetry:} && R_\xi(-\tau) = R_\xi(\tau) && \text{(satisfied)} \\ &\textit{Maximum at origin:} && |R_\xi(\tau)| \le R_\xi(0) = N_0 && \text{(satisfied)} \\ &\textit{Non-negative PSD:} && S_\xi(\omega) = \int_{-\tau'}^{\tau'} N_0 e^{-j\omega\tau}\, d\tau = 2N_0\tau'\,\mathrm{sinc}\left(\frac{\omega\tau'}{\pi}\right) && \text{(violated)} \end{aligned}$$

Since $\mathrm{sinc}(x)$ takes negative values for $|x| > 1$, the PSD in (1.69) is negative at certain frequencies. Hence the rectangular ACF is inadmissible—a conclusion already reached in ?? via Bochner's theorem.

This contrast illustrates a fundamental constraint: while bandlimited PSDs (frequency-domain truncation) yield valid ACFs, time-domain truncation of ACFs yields invalid PSDs.

Pink ($1/f$) noise

Pink noise, also called $1/f$ noise or flicker noise, is commonly modeled by a PSD that decays inversely with frequency over a finite band:

$$S_{\text{pink}}(\omega) = \begin{cases} \dfrac{K}{|\omega|}, & \omega_L \le |\omega| \le \omega_H, \\ 0, & \text{otherwise}, \end{cases} \tag{1.70a}$$

$$P_{\text{pink}} = R_{\text{pink}}(0) = \frac{1}{2\pi} \int_{-\infty}^{\infty} S_{\text{pink}}(\omega)\, d\omega = \frac{K}{\pi} \ln\left(\frac{\omega_H}{\omega_L}\right), \tag{1.70b}$$

$$R_{\text{pink}}(\tau) = \frac{K}{\pi} \left[\mathrm{Ci}(\omega_H|\tau|) - \mathrm{Ci}(\omega_L|\tau|)\right], \qquad \tau \neq 0. \tag{1.70c}$$

At the origin, $R_{\text{pink}}(0)$ is understood by continuity and equals P_{pink}.

Here, $K > 0$ is a scaling constant, $0 < \omega_L < \omega_H$, and

$$\mathrm{Ci}(x) = -\int_x^{\infty} \frac{\cos u}{u}\, du$$

is the cosine integral.

For large time lags, the asymptotic expansion

$$\mathrm{Ci}(x) \sim \frac{\sin x}{x} - \frac{\cos x}{x^2} + \mathcal{O}(x^{-3})$$

$$\text{gives} \quad R_{\text{pink}}(\tau) \sim \frac{K}{\pi|\tau|} \left[\frac{\sin(\omega_H|\tau|)}{\omega_H} - \frac{\sin(\omega_L|\tau|)}{\omega_L}\right], \qquad |\tau| \to \infty,$$

up to terms of order $\mathcal{O}(|\tau|^{-2})$. Thus, finite-band pink noise exhibits slow oscillatory correlation decay, with the low-frequency edge dominating the large-lag envelope when $\omega_L \ll \omega_H$.

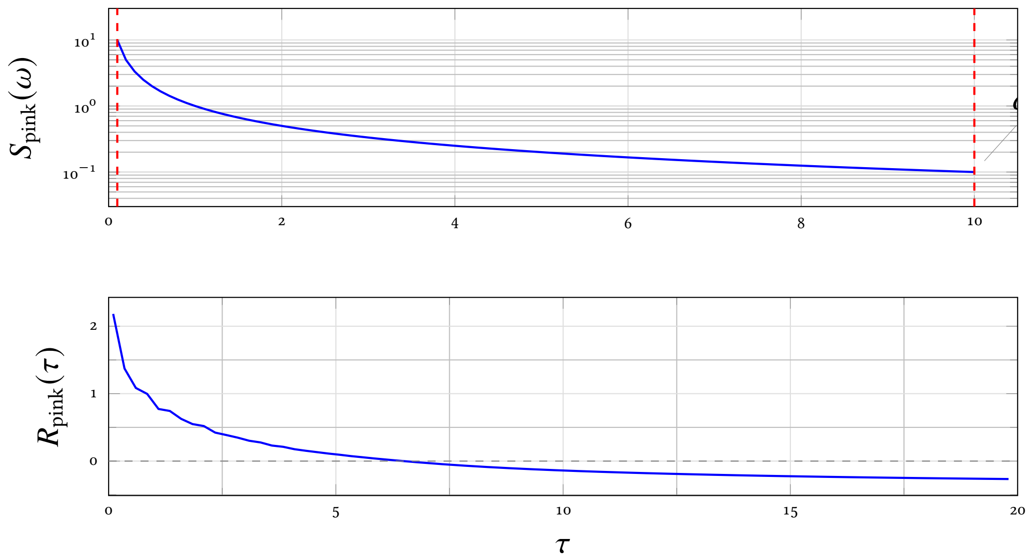


Figure 1.12: Pink noise: (top) $1/\omega$ PSD with cutoffs $\omega_L = 0.1$ and $\omega_H = 10$; (bottom) exact ACF computed from $R(\tau) = \frac{K}{\pi}\left[\mathrm{Ci}(\omega_H\tau) - \mathrm{Ci}(\omega_L\tau)\right]$, showing slow $1/\tau$ decay and long-range memory.

Linear Mixture Models

Linear mixture models provide a flexible framework for constructing non-Gaussian or multi-modal power spectra from elementary Gaussian components. We consider the weighted superposition

$$r_x(\lambda) = \sum_{k=-K}^{K} c_k\, \phi(\lambda; \mu_k, \alpha), \qquad \phi(\lambda; \mu, \alpha) \triangleq e^{-\alpha(\lambda-\mu)^2}, \tag{1.71}$$

where $\alpha > 0$ controls the component width and the weights are

$$c_k = \begin{cases} \sigma^2, & k = 0, \\ \dfrac{\sigma^2}{|k|}, & k \neq 0. \end{cases} \tag{1.72}$$

Because $\sum_{k\neq 0} |c_k|$ diverges like the harmonic series, K must remain finite. In the limit $K \to \infty$ the frequency-domain mixture would approach a Dirac comb, which is not an absolutely integrable PSD.

Case 1: Time-lag domain $(\lambda = \tau)$. Here $R_x(\tau) = r_x(\tau)$ is a candidate ACF.

- *Zero-mean components* $(\mu_k = 0)$. All Gaussians are centered at the origin, so

$$R_x(\tau) = \Big(\sum_{k=-K}^{K} c_k\Big) e^{-\alpha\tau^2}.$$

For $K = 0$ this reduces to $R_x(\tau) = \sigma^2 e^{-\alpha\tau^2}$, whose PSD is the Fourier pair

$$S_x(\omega) = \sigma^2 \sqrt{\frac{\pi}{\alpha}}\, e^{-\omega^2/(4\alpha)}.$$

Because the PSD is a single positive Gaussian, $R_x(\tau)$ is a valid ACF.

- *Shifted components* $(\mu_k = k)$. The candidate ACF becomes

$$R_x(\tau) = \sigma^2 e^{-\alpha\tau^2} + \sum_{k=1}^{K} \frac{\sigma^2}{k} \Big[e^{-\alpha(\tau-k)^2} + e^{-\alpha(\tau+k)^2}\Big],$$

which is even and maximal at the origin. Its Fourier transform, however, is

$$S_x(\omega) = \sigma^2 \sqrt{\frac{\pi}{\alpha}}\, e^{-\omega^2/(4\alpha)} \Big[1 + 2\sum_{k=1}^{K} \frac{\cos(k\omega)}{k}\Big].$$

The bracketed factor is the partial sum of a Fourier series that becomes negative for certain ω when K is large. Hence K must be kept small enough that $S_x(\omega) \geq 0$ for all ω.

Case 2: Frequency domain $(\lambda = \omega)$. Here $S_x(\omega) = r_x(\omega)$ is a candidate PSD. All transforms use the angular-frequency convention $S(\omega) = \int R(\tau) e^{-j\omega\tau}\, d\tau$ with $R(\tau) = \frac{1}{2\pi} \int S(\omega) e^{j\omega\tau}\, d\omega$.

- *Zero-mean components* ($\mu_k = 0$). For $K = 0$,

  $$S_x(\omega) = \sigma^2 e^{-\alpha\omega^2},$$

  whose inverse Fourier transform gives the ACF

  $$R_x(\tau) = \frac{\sigma^2}{2\sqrt{\pi\alpha}} e^{-\tau^2/(4\alpha)}.$$

  The total power is $R_x(0) = \sigma^2/(2\sqrt{\pi\alpha})$.
- *Shifted components* ($\mu_k = k$). The PSD is a sum of shifted, positive Gaussian kernels:

  $$S_x(\omega) = \sigma^2 e^{-\alpha\omega^2} + \sum_{k=1}^{K} \frac{\sigma^2}{k} \left[e^{-\alpha(\omega-k)^2} + e^{-\alpha(\omega+k)^2} \right].$$

  Because every term is non-negative, $S_x(\omega) \geq 0$ automatically; no constraint on K is required for positive definiteness. The corresponding ACF is

  $$R_x(\tau) = \frac{\sigma^2}{2\sqrt{\pi\alpha}} e^{-\tau^2/(4\alpha)} \left[1 + 2\sum_{k=1}^{K} \frac{\cos(k\tau)}{k} \right].$$

  Properties:

  - *Envelope:* Gaussian decay $\dfrac{\sigma^2}{2\sqrt{\pi\alpha}} e^{-\tau^2/(4\alpha)}$.
  - *Modulation:* Harmonic series $1 + 2\displaystyle\sum_{k=1}^{K} \frac{\cos(k\tau)}{k}$.
  - *Peak value:* $R_x(0) = \dfrac{\sigma^2}{2\sqrt{\pi\alpha}} \big(1 + 2H_K\big)$, where $H_K = \sum_{k=1}^{K} \frac{1}{k} \approx \ln K + \gamma$ is the Kth harmonic number.
  - *Long-lag behavior:* The envelope decays as $e^{-\tau^2/(4\alpha)}$, but the harmonic modulation persists, creating side-lobe structure in the ACF.

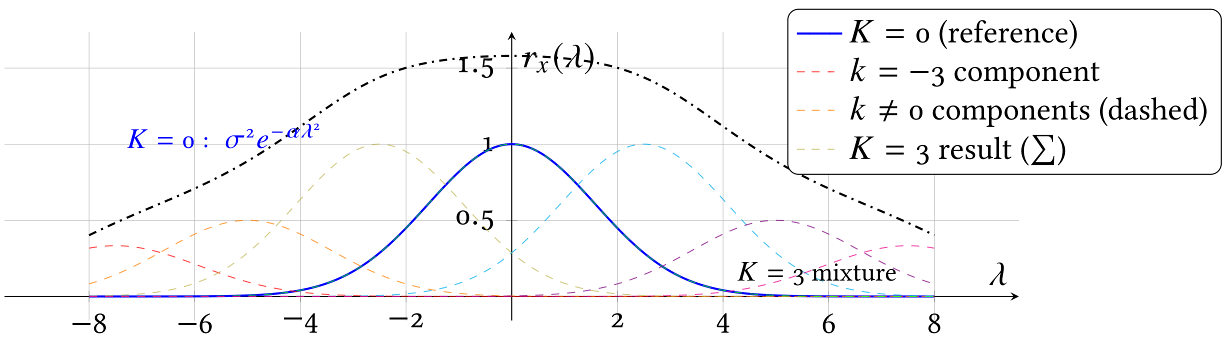


Figure 1.13: Gaussian Mixture Model: $K = 0$ vs $K = 3$

Example: Find the number of components needed in Equation (1.71) for modeling white Gaussian noise.
Consider the Gaussian Mixture Model restricted to the k=0 component:

$$r_x(\lambda) \approx \lim_{\alpha \to \infty} \sigma^2 \cdot \mathcal{N}\left(\lambda; \mu_0, \frac{1}{\sqrt{2\alpha}}\right) = \delta(\lambda)$$

where $\mu_0 = 0$ (centered at lag/frequency origin).

Table 1.1: Practical Guidelines for Component Selection in GMM-Based Noise Modeling

Scenario	**Reason**	**Typical K**
Finite α	Real systems cannot achieve $\alpha \to \infty$; multiple narrow Gaussians better approximate $\delta(\lambda)$	1–3
Numerical stability	Very small variance Gaussians ($\sigma_g \ll 1$) cause computational issues (underflow, ill-conditioning)	2–5
Bandlimited approximation	Modeling white noise over finite bandwidth $[-B, B]$ requires components spanning the support	$K \approx \left\lceil \frac{2B}{\Delta\mu} \right\rceil$

The above convergence to the Dirac delta function yields the following autocorrelation function:

$$r_x(\lambda) \to \sigma^2 \delta(\lambda)$$

This exactly reproduces the autocorrelation of White Gaussian Noise. Note that adding components with $k \neq 0$ introduces frequency-dependent structure, violating the independence assumption, and generating models of colored (non-white) spectrum.
While one component ($K = 0$) is theoretically sufficient for modeling ideal White Gaussian Noise, practical implementations often require additional components. The table below summarizes common scenarios:
Rule of Thumb for Finite α. Let the Gaussian component width be defined as $\sigma_g = \frac{1}{\sqrt{2\alpha}}$. The effective support of a Gaussian is approximately $\pm 3\sigma_g$ (covering ~99.7% of its mass). To approximate the Dirac delta $\delta(\lambda)$ within a tolerance ε, the number of components should satisfy:

$$K_{\text{practical}} \approx \left\lceil \frac{3/\sqrt{2\alpha}}{\Delta\mu} \right\rceil \tag{1.73}$$

where $\Delta\mu$ is the spacing between adjacent component means μ_k, $\lceil \cdot \rceil$ denotes the ceiling function (round up to nearest integer), and the numerator $3/\sqrt{2\alpha} = 3\sigma_g$ represents the half-width of the effective support.

522

Narrowband Random Signals

Given a complex baseband random signal $x(t)$, its frequency-shifted (modulated) version is defined as

$$y(t) = x(t)e^{j\omega_c t}, \tag{1.74}$$

where ω_c denotes the carrier *angular* frequency. Using the polar decomposition $x(t) = a(t)\exp\{j\phi(t)\}$, where $a(t){=}|x(t)|$ is the *envelope* and $\phi(t){=}\arg\{x(t)\}$ is the baseband *phase*, one obtains

$$y(t) = a(t)e^{j(\omega_c t + \phi(t))}. \tag{1.75}$$

Sample realizations of $y(t)$ exhibit the structure of a modulated harmonic oscillation, commonly represented in canonical form as

$$\operatorname{Re}\{y(t)\} = a(t)\cos\big(\omega_c t + \phi(t)\big), \quad a(t) \geq 0, \quad -\pi < \phi(t) \leq \pi, \tag{1.76a}$$

$$= a_c(t)\cos(\omega_c t) - a_s(t)\sin(\omega_c t), \tag{1.76b}$$

$$\text{where} \quad a_c(t) = a(t)\cos\phi(t), \quad a_s(t) = a(t)\sin\phi(t). \tag{1.76c}$$

$$\text{Then} \quad a(t) = \sqrt{a_c^2(t) + a_s^2(t)}, \quad \phi(t) = \arg\{a_c(t) + j a_s(t)\}, \tag{1.76d}$$

Hence, wherever $a(t){\neq}0$ and the phase is differentiable, the *instantaneous angular frequency* is defined as

$$\begin{aligned} \omega_i(t) &= \dot{\theta}(t) = \omega_c + \dot{\phi}(t) \\ &= \omega_c + \frac{\dot{a}_s(t)a_c(t) - \dot{a}_c(t)a_s(t)}{a_c^2(t) + a_s^2(t)}, \end{aligned} \tag{1.77}$$

where the overdot denotes differentiation with respect to time.

The representations in Equations (1.76a) and (1.76b) admit a geometric interpretation: the envelope $a(t)$ is the magnitude of a vector whose projections onto the axes of a rectangular coordinate system are $a_c(t)$ and $a_s(t)$, and the phase $\phi(t)$ is the angle subtended between the abscissa axis and the vector direction. Both $a(t)$ and $\phi(t)$ vary randomly in time, causing the tip of the vector to execute a random walk in the plane; the narrowband process $y(t)$ can thus be visualized as a rotating vector with randomly fluctuating amplitude and phase. The projections $a_c(t)$ and $a_s(t)$ are accordingly termed the quadrature components of $y(t)$.

Assuming that $x(t)$ in Equation (1.74) is a *zero-mean*, complex, wide-sense stationary process with ACF $R_x(\tau)$ and PSD $S_x(\omega)$, the modulated process $y(t)$ remains wide-sense stationary, and its second-order statistics satisfy [2]:

$$R_y(\tau) = R_x(\tau)e^{j\omega_c \tau}, \tag{1.78a}$$

$$S_y(\omega) = S_x(\omega - \omega_c). \tag{1.78b}$$

Therefore, frequency shifting induces a linear phase rotation in the ACF and a pure translation of the PSD, without altering its shape.

Narrowband Random Signals. Any zero-mean stationary process $y(t)$ is called *narrowband* if the angular bandwidth $\Delta\omega$ of the frequency region where the PSD $S_y(\omega)$ is practically nonzero is small compared to some central angular frequency ω_c of this region, i.e.,

$$S_y(\omega) \neq 0 \quad \text{for } \omega_c - \Delta\omega/2 < \omega < \omega_c + \Delta\omega/2, \quad \Delta\omega \ll \omega_c, \tag{1.79}$$

where $S_y(\omega)$ denotes the one-sided PSD, and $\Delta\omega$ is the effective angular bandwidth, which can be defined in various ways (e.g., 3-dB bandwidth, RMS bandwidth, or equivalent noise bandwidth).

If the one-sided spectral density $S_y(\omega)$ is symmetric with respect to some angular frequency ω_c, then it is natural to select this frequency as the central frequency. In other cases, ω_c may be chosen more or less arbitrarily. A common choice is the spectral centroid (i.e., the first moment of the normalized spectral density):

$$\omega_c = \int_0^\infty \omega\, S_y(\omega)\, d\omega \Big/ \left(\int_0^\infty S_y(\omega)\, d\omega \right) \tag{1.80}$$

The ACF of a narrowband random process can always be represented in the form :

$$\begin{aligned} R_y(\tau) &= \sigma_y^2 \big(r_c(\tau)\cos(\omega_c\tau) - r_s(\tau)\sin(\omega_c\tau) \big) \\ &= \sigma_y^2\, r(\tau)\cos\big(\omega_c\tau + \gamma(\tau)\big), \end{aligned} \tag{1.81a}$$

$$\text{where } r(\tau) = \sqrt{r_c^2(\tau) + r_s^2(\tau)}, \quad \tan\gamma(\tau) = \frac{r_s(\tau)}{r_c(\tau)}, \tag{1.81b}$$

$$\sigma_y^2 r_c(\tau) = \frac{1}{2\pi} \int_{-\omega_c}^\infty S_y(\omega_c + \nu)\cos(\nu\tau)\, d\nu, \tag{1.81c}$$

$$\sigma_y^2 r_s(\tau) = \frac{1}{2\pi} \int_{-\omega_c}^\infty S_y(\omega_c + \nu)\sin(\nu\tau)\, d\nu, \tag{1.81d}$$

with the initial conditions $r_c(0){=}1, r_s(0){=}0, \gamma(0){=}0$. Here, $\sigma_y^2{=}R_y(0)$ denotes the variance of the narrowband process $y(t)$.

The functions $r_c(\tau)$ and $r_s(\tau)$ are respectively the *in-phase* and *quadrature* correlation components. The envelope $r(\tau)$ and phase modulation $\gamma(\tau)$ characterize the slow variations of the narrowband process relative to the carrier frequency ω_c.

In the particular case where the spectral density $S_y(\omega)$ is symmetric with respect to the central angular frequency ω_c, the ACF of the narrowband process simplifies to

$$R_y(\tau) = \sigma_y^2\, r(\tau)\cos(\omega_c\tau) \simeq \frac{\cos(\omega_c\tau)}{2\pi} \int_{-\infty}^\infty S_y(\omega_c + \nu)\cos(\nu\tau)\, d\nu, \tag{1.82}$$

since $r_s(\tau){=}0,\ \gamma(\tau){=}0$.

Because the right-hand sides of Equations (1.81c) and (1.81d) involve the spectral density of the process $y(t)$ shifted to baseband (i.e., the low-frequency spectrum), the functions $r_c(\tau)$, $r_s(\tau)$, as well as $r(\tau)$ and $\gamma(\tau)$, vary slowly compared to $\cos(\omega_c\tau)$.

Task. Show that if $y(t)$ is ruled by a Gaussian distribution, then

- $A_c(t)$ and $A_s(t)$ are also Gaussian-distributed, each one having variance $\sigma_c^2 = \sigma_s^2 = \sigma_y^2$
- $A_c(t)$ and $A_s(t)$ are zero-mean, if $y(t)$ is zero mean,
- $A_c(t)$ and $A_s(t)$ are zero-correlated for all lag values τ, if $S_y(\omega)$ is symmetric with respect to ω_c
- For a zero-mean Gaussian narrowband process $y(t)$, the envelope $A(t)$ is Rayleigh-distributed with variance σ_y^2.

Cosine-Modulated Lorentzian ACF Model. Consider the following candidate ACF:

$$R_y(\tau) = \sum_{k=-K}^{K} w_k e^{-\alpha|\tau|} \cos(k\omega_c \tau), \tag{1.83a}$$

$$w_k = \frac{\sigma^2}{|k| + (1 - \operatorname{sgn}(|k|))}, \tag{1.83b}$$

where $\sigma^2 > 0$, ω_c is an angular frequency parameter, and $\alpha > 0$ controls the decay rate of each component.

To show that $R_y(\tau)$ is a valid ACF, we verify the following properties:

(i) *Hermitian symmetry.* Since $R_y(\tau)$ is real-valued, and since $|-\tau| = |\tau|$ and $\cos(-\theta) = \cos(\theta)$, it follows that $R_y(-\tau) = R_y(\tau) = R_y^*(\tau)$.

(ii) *Maximum at the origin.* At $\tau = 0$,

$$R_y(0) = \sum_{k=-K}^{K} w_k.$$

For $\tau \neq 0$,

$$|R_y(\tau)| = e^{-\alpha|\tau|} \left| \sum_{k=-K}^{K} w_k \cos(k\omega_c \tau) \right| \leq e^{-\alpha|\tau|} \sum_{k=-K}^{K} w_k < R_y(0),$$

since $\alpha > 0$ and $w_k > 0$.

(iii) *Non-negative power spectral density.* The Fourier transform of $e^{-\alpha|\tau|}$ is

$$G(\omega) = \frac{2\alpha}{\alpha^2 + \omega^2} \geq 0.$$

Using the Fourier modulation property, the spectral contribution of the k-th term is

$$S_k(\omega) = \frac{w_k}{2} \left(G(\omega - k\omega_c) + G(\omega + k\omega_c)\right) \geq 0.$$

Hence, $S_y(\omega) = \sum_{k=-K}^{K} S_k(\omega) \geq 0.$

Therefore, by the Wiener–Khintchine theorem, $R_y(\tau)$ is positive semi-definite and thus a valid autocorrelation function.

Consequently, the model satisfies Hermitian symmetry, is bounded with maximum at the origin, and has a non-negative power spectral density. Therefore, $R_y(\tau)$ is a valid autocorrelation function.

For $K = 1$, one obtains

$$R_y(\tau) = \sigma^2 e^{-\alpha|\tau|}\left(1 + 2\cos(\omega_c \tau)\right),$$
$$S_y(\omega) = 2\alpha\sigma^2 \left(\frac{1}{\alpha^2 + \omega^2} + \frac{1}{\alpha^2 + (\omega - \omega_c)^2} + \frac{1}{\alpha^2 + (\omega + \omega_c)^2}\right),$$

which corresponds to three Lorentzian peaks centered at $\omega = 0$ and $\omega = \pm\omega_c$.

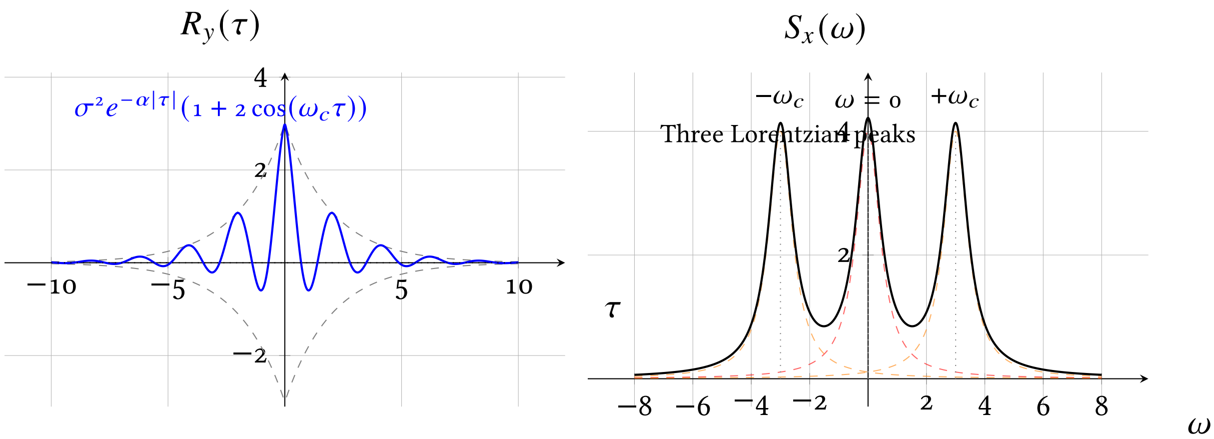


Figure 1.14: the PSD of two zero-mean random processes with ACF corresponding to k=1.

Pytask: Lab: Envelope modulation of Random Signals

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